Carrying Immersion over Networks

A Systematic Review of Networked Extended Reality, 2016–2026

Networked XR Atlas

October 2026

Abstract

Extended reality (XR) asks more of a network than any earlier medium. It needs hundreds of megabits per second for 360-degree and volumetric content, a motion-to-photon deadline of a few tens of milliseconds, a continuous uplink of pose and sensor data, and a world that several people share. This review maps the research on meeting these demands. It draws on the Networked XR Atlas: 5,408 papers published between October 2016 and October 2026, screened from 39,476 OpenAlex records, labelled on five dimensions and read in depth where they mattered most.

We organise the field by the six things a networked XR system must do: *encode* the scene, *anticipate* what the user will need, decide where to *compute*, *provision* the network, *connect* people and machines, and *assure* experience and trust. For each function we report what is established, what is contested and what is merely assumed.

Three findings stand out. First, the field has moved from delivering content to allocating resources. Work on anticipation (adaptive streaming, prediction, caching) fell from 27.5% to 17.4% of output between 2017–19 and 2022–24, while work on provisioning the network rose from 14.6% to 27.2%. Second, evidence has weakened as the field has grown. Evaluation on real networks or testbeds fell from 38.1% to 28.2% of papers. Over the same period, simulation and analysis rose from 26.2% to 39.4%, and only 7% of learning-based papers were tested on real networks. Third, network metrics are seldom tied to what users perceive. Requirement figures are copied from survey to survey rather than measured, and the uplink, multi-user scale and the cost of security are rarely studied. We close with nine cross-cutting gaps and a research agenda ordered by what each project unlocks for the others.

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1 Introduction

A head-mounted display shows a world that must respond to the head that wears it. When the user turns, the new view must appear within a few tens of milliseconds, or the image lags the body and comfort fails [1]. On a standalone device this deadline is met locally. Once the scene, the computation or the other participants are elsewhere, the network becomes part of the perceptual loop. Extended reality (XR), the umbrella term for virtual, augmented and mixed reality, therefore asks more of a network than video or the web. A 360-degree or volumetric stream needs hundreds of megabits per second if it is sent whole [2, 3]. Remote rendering must finish a frame within the motion-to-photon (MTP) budget [1]. The device sends pose, camera frames and captured media upstream all the time. When several people share a world, the network must also keep them consistent with each other [4].

Ten years of research have produced many answers to these demands, and they are scattered across communities that seldom read each other. Multimedia-systems researchers built viewport-adaptive streaming. Mobile-computing researchers offloaded AR recognition to the edge [5]. Wireless researchers framed XR as a joint allocation of radio, computing and energy. Haptics researchers pursued the tactile internet [6]. Visionaries placed all of it inside 6G and the metaverse [7, 8]. Existing surveys follow these boundaries. Each covers one medium, such as 360-degree video [2, 9, 10], point clouds [3] or haptics [11], or one technology, such as edge computing [5, 12], ultra-low latency [13] or the 6G metaverse [7, 14]. None measures the field as a whole, and none asks how well its claims are tested.

This review does both. It condenses the *Networked XR Atlas*, a corpus of 5,408 papers published between October 2016 and October 2026. The atlas was screened from 39,476 records with a validated procedure, labelled on five dimensions and read in depth where it mattered most (Section 2). The atlas supports three kinds of statement: counts and trends over the whole corpus, syntheses of the papers that shape each topic, and gaps that have been checked against a text search of every included paper.

Organising idea. We do not organise the field by technology or by medium. We organise it by *function*: the six things any networked XR system must do, whatever its medium and access network (Section 4). A system must *encode* the scene into something that can be sent (Section 5). It must *anticipate* what the user will look at or do next, so that the right content arrives in time (Section 6). It must decide where to *compute*, on the device, at the edge or in the cloud, and how to hide the delay this adds (Section 7). It must *provision* the network with radio, scheduling, transport and architecture that supply the rate, delay and reliability required (Section 8). It must *connect* users with each other and with remote machines, through shared state, touch and telepresence (Section 9). And it must *assure* that the result is good enough and safe for the people who use it (Section 10). Each function poses its own design question, and the answers can be compared on the same terms.

Contributions. The review makes four contributions.

(1) A **functional taxonomy** of networked XR that places the 15 networking problems and 9 content types of the atlas codebook into six functions, each defined by a design question (Section 4, Figure 4 and Table 1).

(2) A **quantitative account of how the field has changed**: its growth, the move from delivering content to allocating resources, the replacement of 360-degree video by generic XR traffic as the reference workload, and the decline of evaluation on real systems (Section 3).

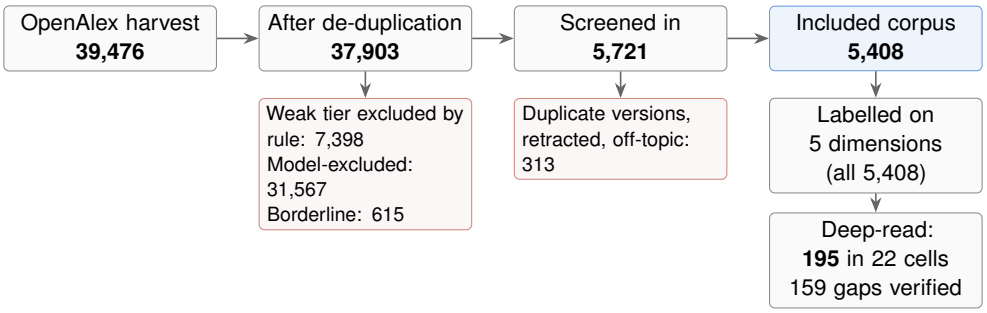


Figure 1: Flow of records from harvest to analysis. Excluded-record counts are not additive with the included count, because borderline records are kept apart and screening ran before the second de-duplication pass.

(3) A **synthesis of each function** that separates what is established, what is contested and what is assumed, with the measured results that support each claim (Sections 5–10).

(4) **Nine cross-cutting gaps** and a research agenda ordered by dependency: which projects produce the data and models that the others need (Sections 11 and 12).

2 Review Method

The review rests on the Networked XR Atlas, whose full protocol, code and intermediate decisions are public. Here we summarise the steps that bear on how the evidence should be read. Figure 1 shows the flow of records.

Research questions. (RQ1) How large is the networked-XR literature, and how has its composition changed? (RQ2) What does it establish about each function of a networked XR system, and with what evidence? (RQ3) How realistically are its claims tested? (RQ4) Which gaps recur across functions, and which research would close them first?

Search. We queried OpenAlex, which indexes journals, proceedings, theses, books and preprints. A record qualified if its title or abstract contained one of 20 XR phrases *and* one of 47 networking phrases. The XR side covers virtual, augmented, mixed and extended reality, the metaverse, 360-degree and volumetric video, holographic communication, telepresence and the tactile internet. The networking side runs from streaming, edge computing and remote rendering to 5G, 6G, Wi-Fi, QoE and caching. The XR side deliberately includes media terms. A search for "virtual reality" alone would miss much of the field, because many papers describe their content only as haptic traffic, omnidirectional video or point clouds. The window runs from October 2016 to October 2026. After removing 1,573 duplicates by identifier, DOI and title, 37,903 records remained.

Screening. Records were sorted into tiers by how strongly XR and networking terms co-occur. In a hand-labelled pilot of 100 records, the weakest tier held no relevant paper, so its 7,398 records were excluded by rule. The other 30,472 records were screened by a calibrated classification model against a written scope. A paper was included when transmitting, offloading, rendering remotely, caching, synchronising or securing XR content *over a network* was its core or a substantial subject. On the decidable pilot records, the inclusion rule reached precision 0.84 and recall 0.94. A second de-duplication pass merged preprints with their published versions, and retracted and off-topic records were removed (313 in all). The final corpus holds 5,408 papers: 48% conference papers, 40% journal articles, and the rest preprints, theses and chapters.

Labelling. Every included paper was labelled on five closed dimensions, each with written definitions: the *networking problem* it mainly addresses (15 options), the *network technology* (11), the

content type transmitted (9), the *method* (10) and the *evaluation* (6). 3,320 papers were labelled by the model and the remaining 2,088 by direct reading of title and abstract. On 160 papers labelled both ways, the primary labels agree for 59%–86% of papers. The top-two label sets overlap for 87%–97%, so most disagreements concern papers that sit between two categories. The labels support counts and trends over hundreds of papers. They do not support claims about any single paper, and every paper discussed individually below was read.

Evaluation realism as evidence strength. Reviews of interventions grade evidence by study design. In networking the main threat to a claim is an unrealistic test environment, so we grade evidence by *evaluation realism*, from most to least realistic: a real network or testbed with human users; a real network or testbed without users; emulation driven by real traces; simulation; analysis alone; and no evaluation. Abstracts often blur "prototype" and "emulation", so this is the least reliable dimension (top-1 agreement 63% against 80% for content). Differences of a few percentage points in realism should not be over-read.

Deep reading and gap verification. The atlas divides the corpus into 22 cells, one per networking problem and one per specific content type. In each cell it selected the most influential papers, by citations, citations per year and in-corpus PageRank, together with the strongest recent papers and the main surveys. In all, 195 papers were read and summarised in a fixed schema, 120 of them in full text. A candidate gap was kept only if it passed three tests. It had to be meaningful rather than definitionally empty. It had to survive a regular-expression search of every included abstract, which corrects for the single-label taxonomy. And the papers that had been read had to leave it open. 159 gaps passed, 57 of them with high confidence. The open problems that close Sections 5–10 are drawn from them.

Threats to validity. OpenAlex lacks an abstract for 7.0% of the included papers. Work that uses none of our XR phrases is missed. The pilot contained only 17 relevant papers, so precision and recall carry wide intervals. A single-label taxonomy undercounts papers that span categories, so category sizes count the papers for which a category is *primary*. Selecting papers for deep reading by citation favours older work, and a recent-paper quota corrects this only in part. Text-search counts depend on how authors word their abstracts, so they are indicative to within a few points. Screening and labelling used a language model, and drafts were checked against the source texts. Every count in this review is generated by script from the atlas data files.

3 The Field at a Glance

Growth. Annual output rose from 213 papers in 2017 to 828 in 2024, a factor of 3.9 and a compound growth of 21% a year (Figure 2a). Growth was steady until 2022 and then jumped, largely with the metaverse. Papers that use the word went from 2 in 2021 to 79 in 2022 and 268 in 2023. More than half of all included papers (56%) appeared from 2023 onward. The field is conference-led: 48% of papers are conference papers. It is also spread across communities that seldom publish in each other's venues. Communications venues (Globecom, ICC, *IEEE Network*, *JSAC*) sit beside multimedia-systems venues (MMSys, ACM Multimedia, *IEEE Transactions on Multimedia*) and broad open-access journals. The ten largest venues hold only 16% of the corpus. XR venues proper barely appear, so perceptual results from the XR community, such as tolerable latency, reach networking papers late.

From delivering content to allocating resources. The composition of the field changed more than its size (Figure 3). In 2017–19 the largest problem was rate adaptation and streaming, with 23.1% of papers. By 2022–24 its share had fallen to 12.4%, the largest decline of any problem.

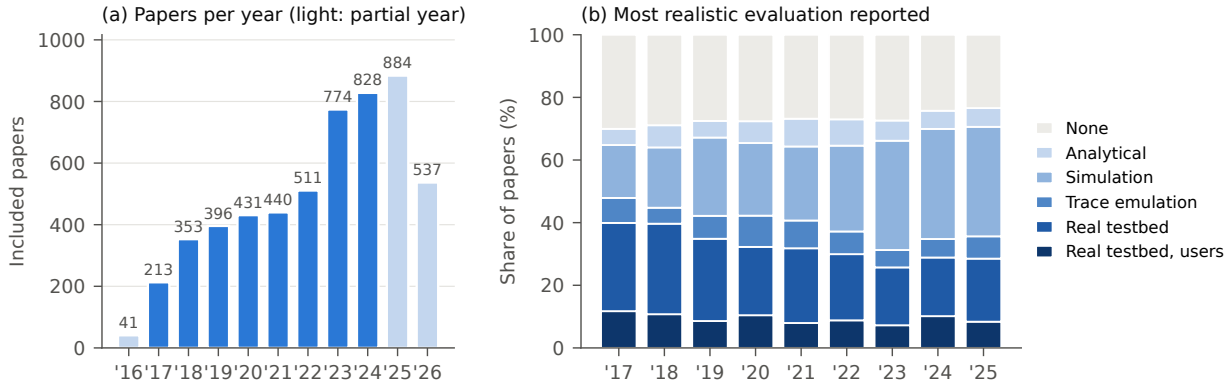


Figure 2: Growth and evaluation realism. (a) Included papers per year; 2016, 2025 and 2026 are partial because of the harvest window and indexing lag. (b) Share of papers by the most realistic evaluation they report. Evaluation on real networks or testbeds (the two darkest bands) shrank as simulation grew.

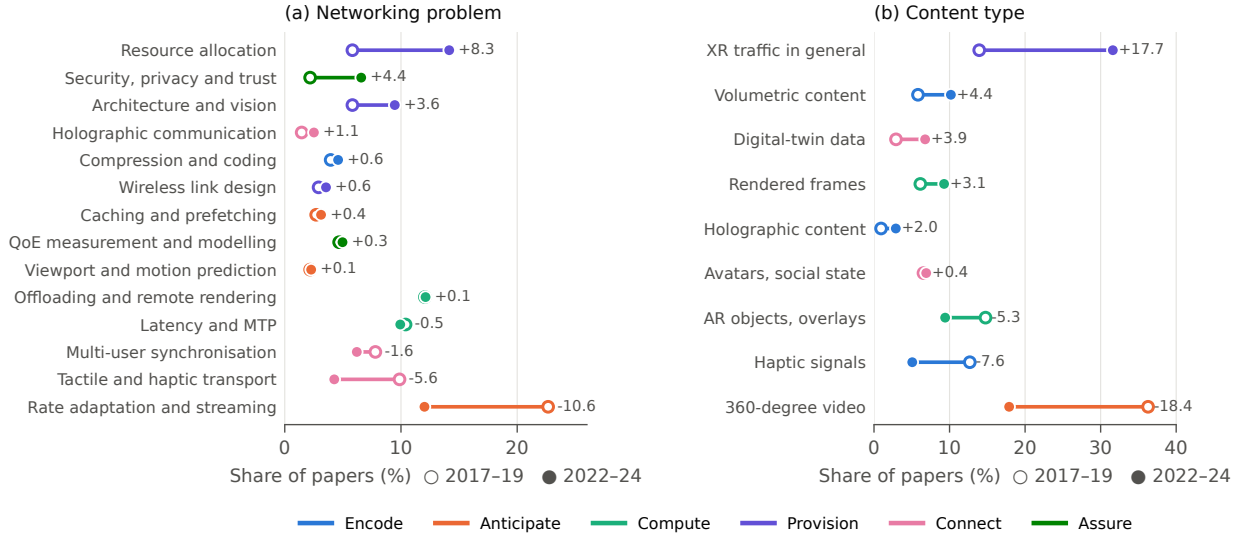


Figure 3: Change in composition between 2017–19 (open circles) and 2022–24 (filled circles), with the difference in percentage points. Colours give the function of Section 4. The field moved from anticipation (streaming 360-degree video) to provisioning (allocating resources for generic XR traffic).

Resource allocation rose from 5.9% to 13.7%, and architecture and vision papers from 6.1% to 9.4%. Security, privacy and trust grew fastest of all, at 48% a year. At the level of functions, anticipation fell from 27.5% to 17.4% of output and provisioning rose from 14.6% to 27.2%. These are shares: streaming still grew in absolute terms, but far more slowly than the field. The shift reflects a change of authors. The early literature was led by multimedia-systems groups building 360-degree players. The later literature is led by wireless groups who formulate XR delivery as a joint allocation of radio, computing and energy, and by authors of 6G and metaverse visions.

From 360-degree video to "XR traffic". The reference workload changed with the problems. 360-degree video was the content of 36.3% of papers in 2017–19 and of 18.8% in 2022–24. The largest gain went to papers that name no specific content and model XR as a rate and a delay bound (13.8% → 30.8%). Among specific contents, volumetric media, digital-twin data, rendered frames and holograms grew, while AR overlays and haptic signals declined. The rise of generic traffic matters for the evidence. A constant-rate flow with a delay bound cannot express viewport dependence, frame-size variability or the coupling between uplink pose and downlink content. These are exactly the properties that the content-specific literature has measured.

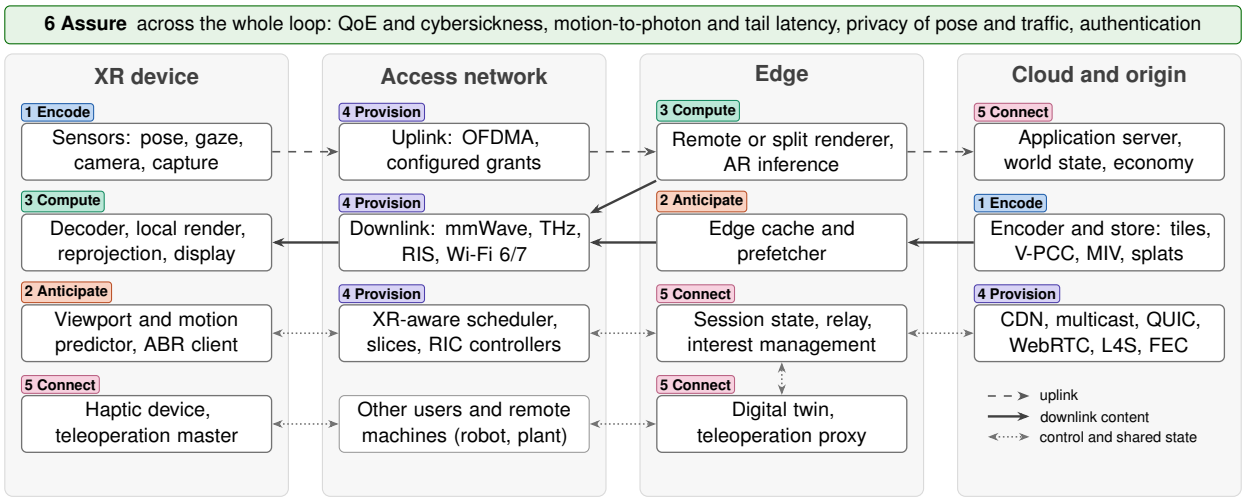


Figure 4: Reference architecture of a networked XR system, with the six functions placed where they act. Rows read left to right as the uplink of pose and captured media, right to left as the downlink of content, and as control and shared state that couple users and remote machines. The same function often acts at several tiers: anticipation runs both on the device (prediction) and at the edge (caching), and computation is split between device and edge.

From systems to learning, and from testbeds to simulators. System implementations fell from 42.3% to 28.8% of papers. Reinforcement learning grew from 1.6% to 9.1%, at 88% a year, and supervised deep learning from 2.4% to 7.2%. User studies, rare to begin with, fell from 4.2% to 2.6%, and in ten years the corpus contains only 37 papers whose main contribution is a dataset. Evaluation followed the methods (Figure 2b). Evaluation on a real network or testbed fell from 38.1% of papers to 28.2%, while simulation and analysis alone rose from 26.2% to 39.4%. The fastest-growing methods are those least often tested on real links: 77% of reinforcement-learning papers are evaluated by simulation alone and only 4% on a real testbed. Learned controllers may well work. But most of their reported gains are gains inside a model of the network, usually one with synthetic channels, traffic and user motion.

What is new. Four clusters of author keywords rose fastest between 2018–20 and 2023–25. The first is the metaverse and 6G ("metaverse" from 1 paper to 501; "6G" from 5 to 217). The second is digital twins (6 $\rightarrow$ 153). The third is learning-based networking, including federated learning (3 $\rightarrow$ 67). The fourth is new representations and transmission paradigms: semantic communication (0 $\rightarrow$ 67), Gaussian splatting (0 $\rightarrow$ 23) and holographic communication (2 $\rightarrow$ 42). Much of this new literature consists of visions and surveys. Sections 5–10 ask how much of it has yet met a real network.

4 A Functional Framework for Networked XR

The atlas labels every paper by the networking problem it addresses and by the content it carries. These labels are precise, but 15 problems and 9 content types are too many to reason about together, and they mix levels: "latency" is a property, "caching" a mechanism and "holographic communication" an application. We therefore group them by *function*, the job a part of the system performs in the loop between the user's body and the content they perceive. A function is defined by a design question that any networked XR system must answer, whatever its medium or access network. Figure 4 places the six functions on a reference architecture, and Table 1 lists the atlas categories that each function contains.

The six functions.

Table 1: The six functions, the atlas categories they contain and their evidence profile. Papers: primary problem label. Share: percentage of all papers in 2017–19 and in 2022–24. Real: evaluated on a real network or testbed, with or without users. Sim.: simulation or analysis only.

Function	Networking problems (atlas cells)	Main content types	Papers	Share 17–19 → 22–24	Real	Sim.
Encode	Compression and coding	Volumetric, holographic, haptic	273	4.0% → 4.6%	28%	49%
Anticipate	Rate adaptation and streaming; viewport and motion prediction; caching and prefetching	360-degree video, volumetric	1,079	27.5% → 17.4%	20%	35%
Compute	Offloading and remote rendering; latency	Rendered frames, AR overlays	1,213	22.5% → 22.1%	42%	31%
Provision	Resource allocation; wireless link design; architecture and vision	XR traffic in general	1,187	14.6% → 27.2%	11%	55%
Connect	Multi-user synchronisation; tactile and haptic transport; holographic communication	Avatars, haptic, holographic, digital twins	814	19.1% → 13.0%	45%	20%
Assure	QoE measurement and modelling; security, privacy and trust	All	572	6.9% → 11.5%	37%	31%

Encode (Section 5). *What representation of the scene travels, and how is it compressed?* Projections and tiling of 360-degree video, point clouds and meshes, multi-view video, neural scene representations, holograms and haptic signals.

Anticipate (Section 6). *What will the user need next, and how is it delivered in time?* Viewport-adaptive streaming, head, gaze and 6DoF motion prediction, and caching and prefetching.

Compute (Section 7). *Where does each stage of perception and rendering run, and how is the added delay budgeted and hidden?* Offloading, remote and split rendering, AR inference at the edge, motion-to-photon budgets, reprojection and device energy.

Provision (Section 8). *How do radio, scheduling, transport and architecture supply the rate, delay and reliability required?* Resource allocation, mmWave, terahertz, reconfigurable surfaces and Wi-Fi, congestion control and loss resilience, and 6G and metaverse architectures.

Connect (Section 9). *How are several people and remote machines kept consistent and responsive?* Multi-user synchronisation, avatars, haptic teleoperation, holographic telepresence and digital twins.

Assure (Section 10). *How do we know the result is good enough, and safe?* QoE measurement and modelling, traffic characterisation, and security, privacy and trust.

Three properties make this grouping useful. First, it is *exhaustive*: every networking problem in the atlas codebook except the residual "other" (270 papers) belongs to exactly one function, and every specific content type has a natural home. Second, the functions are *ordered along the loop*. Encoding and anticipation decide what is sent, computation and provisioning decide where and how, and connection and assurance concern the people at the ends. Third, each function has a *characteristic evidence profile* (Table 1). The functions that build systems (compute, connect) are tested on real testbeds far more often than those that optimise models (provision, encode). The same gap therefore recurs in different words across the sections that follow, and Section 11 returns to it.

How functions and content combine. The two labelling dimensions are not independent, and their dependence shows where the field's attention has and has not gone (Figure 5). The strongest associations are definitional and confirm the labels. Rate adaptation is studied mostly for 360-degree video (551 papers against 170 expected under independence), tactile transport for haptic signals, and

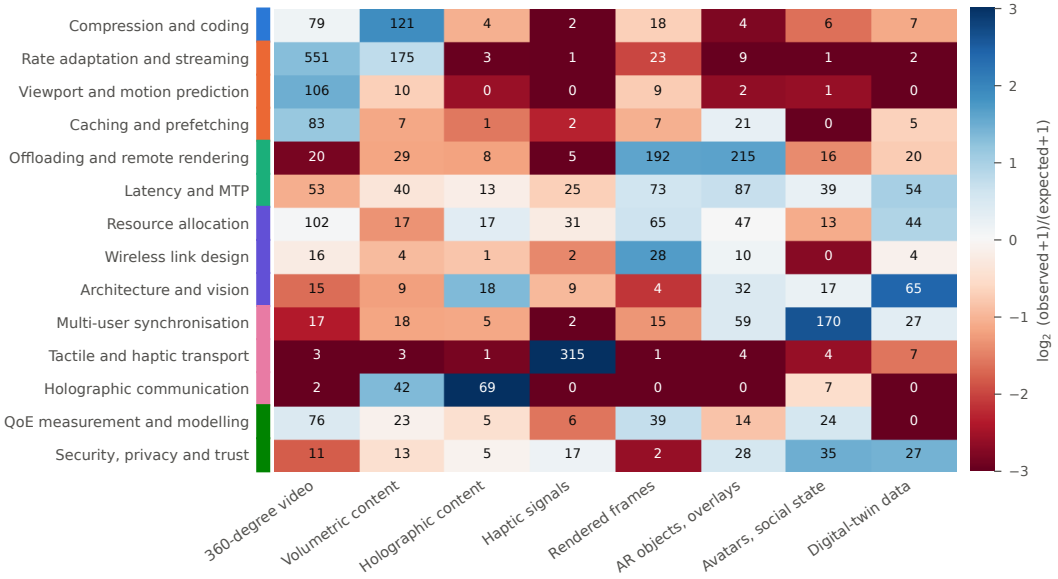


Figure 5: Networking problem (rows, grouped by function) against content type (columns). Numbers are papers. Colour is $\log_2$ of observed over expected counts under independence: blue cells hold more papers than expected and red cells fewer. "XR traffic in general" is omitted.

multi-user synchronisation for avatars. The informative cells are the weak ones. Resource allocation for volumetric content holds 17 papers against 62 expected, and offloading for avatars and social state 16 against 41. QoE work rarely involves an edge deployment (8 against 51), and multi-user synchronisation over 5G is equally rare (12 against 53). The content types that the field now calls its future, volumetric and social, are the ones its allocation and offloading machinery has least often been applied to.

5 What Is Sent: Representing and Encoding Immersive Content

Before a network can schedule, cache or adapt immersive content, a designer must decide what the content is: a sphere of pixels, a set of points, a bundle of camera views, the parameters of a neural scene model or a stream of force samples. This section asks which representation travels over the network and how it is compressed. The coding cell holds 273 papers (5.0% of the corpus), and its share rose from 4.0% in 2017–2019 to 4.6% in 2022–2024. The content it serves has shifted. The share of 360-degree video papers fell from 36.3% to 18.8% over the same periods, that of volumetric content rose from 5.8% to 10.1%, and Gaussian splatting (0 papers, then 23) and semantic communication (0, then 67) appeared from nothing. Evaluation lags the shift: 49% of coding papers are judged by simulation or analysis, 18% not at all and 5% with users. The argument of this section is that the representation fixes three quantities before any network algorithm acts (the bitrate floor, the encoding delay and the decoding cost on the device) and that new representations are proposed faster than these quantities are measured. Figure 6 sets the main chains side by side. Adaptation, prediction and caching are treated in Section 6, and haptic transport and holoportation systems in Section 9.

5.1 360-degree video: a coding problem on a sphere

360-degree video reuses the 2D video chain, and its questions sit at the two ends: how the sphere is mapped to rectangles and how many decoders the client can afford. The established result is that concentrating bits on the viewport saves bitrate, but less than it saves pixels. Oculus's offset cube used fewer than half the pixels of an equirectangular frame for similar view quality, yet saved only 5.6–16.4% of bitrate; its 88 representations (22 orientations $\times$ 4 qualities) made the player

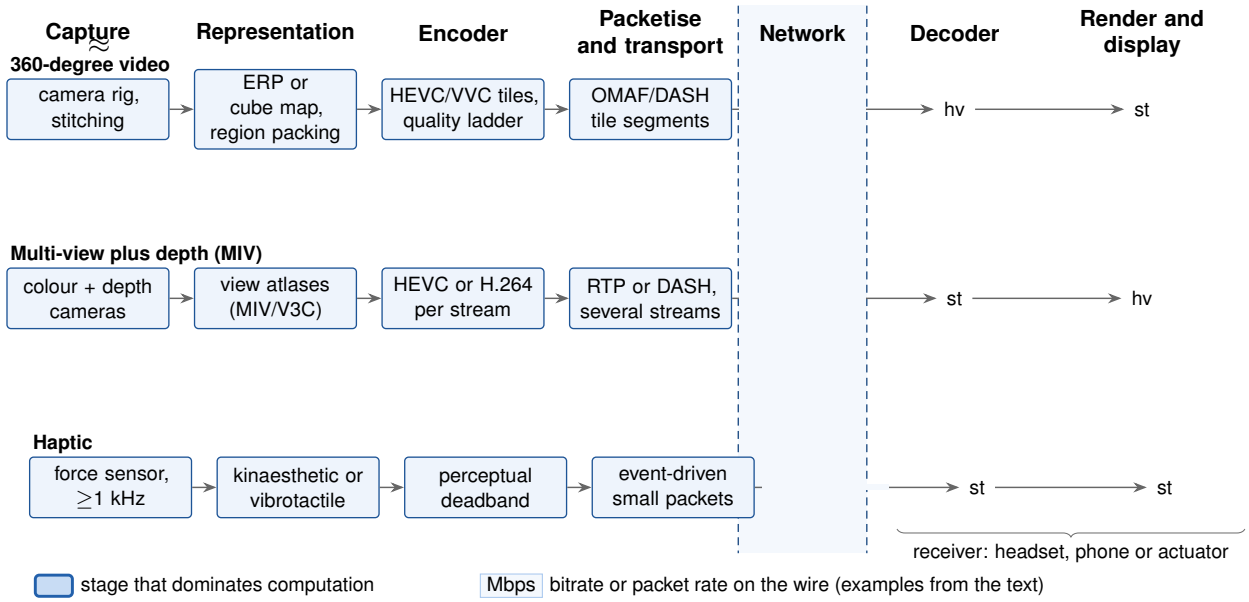


Figure 6: Transmission chains for the five main representations, from capture to display. The bitrate on the wire falls as the representation becomes more abstract, while computation moves to the encoder (V-PCC, per-frame training) or to the receiver (tile decoding, view synthesis, splat rendering). Figures: 360-degree video [15, 16], point clouds [17, 18], multi-view streams [19], Gaussian video [20], deadband coding [21].

download over 57% extra segments [15]. Content-adaptive coding in x265 saved 13–15% BD-rate without subjective loss [22]. At a fixed 6 Mbps a cube map with a full-quality front face beat three other layouts, and pyramids failed under head motion [23]; projections other than equirectangular gain about 10% in coding [24].

The delivery unit is contested. Viewport-specific versions need no reassembly but multiply storage; tiles share one encode but cost 7.3–10.4% in coding efficiency [16], and up to 37.3% for small tiles at low resolution [25]. Point-of-view scalable coding claims about 75% saving [26] against Oculus's 5.6–16.4%, a gap explained mostly by baselines (a whole equirectangular frame against a multi-version ladder). The client pays for tiles: Flare raised phone CPU load from 18% to 58% [16], and streaming took 28.5% of headset power [27], both on 2017–2018 hardware. OMAF standardises projection, region-wise packing and quality descriptors [28], yet network studies almost always assume tiled equirectangular video, so their results may not transfer to deployed formats. Gaze adds a lever with user evidence: gaze-based quantisation saved 40–47% in cloud VR gaming [29], and gaze-contingent compression was five times more effective when motion-to-photon (MTP) latency stayed below 15 ms [30].

5.2 Volumetric content: points, meshes and views

Volumetric content is one to three orders of magnitude heavier than 2D video, and its codecs trade compression ratio against speed. Raw point clouds need 720 Mbit/s at 200K points and 30 fps [18], and about 5 Gbit/s at one million points [17]. V-PCC projects patches onto planes and reuses video codecs; it reaches lossy ratios of 100–500 but took about 4 hours to encode 30 frames of one object [31]. G-PCC and Draco code geometry in 3D, run fast and compress less. Live systems therefore avoid V-PCC. VR2Gather's MPEG Anchor codec cut 281.8 Mbit/s to 19.0 Mbit/s but raised latency from 40.6 to 278.7 ms, and parallel tiles brought it back to 90.6 ms [17]: coding, not transit, set the delay. Tuning the standard codecs yields single-digit gains, such as BD-rate reductions of 2.16% (geometry) and 6.13% (attributes) from V-PCC rate control [32]. Larger gains come from changing the representation: sub-sampling before V-PCC cut bitrate by up to 48.5% at equal quality [33], and 3D patch coding matched V-PCC's ratio with real-time decoding [34].

The receiver is the second constraint. An off-the-shelf 3D super-resolution model ran at about

0.1 fps until it was made 53 times faster on mobile hardware [35], and Hermes reached 30 fps on phones by exploiting inter-frame correlation [36]. No codec wins everywhere: a measurement of Draco, G-PCC, V-PCC and two neural codecs found that the best choice depends on content and viewing behaviour [37], and static mesh codecs show the same trade-off [38].

What deployed live systems send is often views rather than geometry. Project Starline sends four colour and three depth H.265 streams at 30–100 Mbit/s and fuses them at the receiver, because real-time mesh or point-cloud compression was not possible at its resolution [19]; MetaStream codes RGB-D in 2D on the cameras and builds point clouds at the edge [39]. MPEG Immersive Video (MIV) standardises this multi-view-plus-depth path, with atlas packing [40] and view synthesis at the decoder [41], so hardware video decoders do most of the work. Yet only one of about twenty corpus papers on MIV uses a real network, and none has users. Light fields follow the same pattern: DASH delivery of static light fields [42] and multiple-description coding of light-field video [43] were tested without real links. Whether to send geometry or views is the main open question of this strand. No study compares point clouds, meshes, multi-view video and Gaussian splats at matched perceived quality, bitrate and latency.

5.3 Neural and semantic representations

Neural representations replace explicit geometry with model parameters, and semantic codecs replace pixels with meaning. Both promise large gains; neither has yet faced a variable network. The most deployable design reuses existing hardware: V^3 bakes dynamic Gaussians into 2D videos for H.264 decoders and reaches 32.97 dB PSNR at 0.532 MB per frame and 27 fps on a phone [20]. Its costs show where the work moves. Training takes about a minute per frame on a desktop GPU, so live content is out of reach, and download was measured only on an internal network. Other designs offer single-model variable bitrate for NeRF video [44], training 300 times faster than NeRF [45], full-scene splat streaming at 30 fps on phones [46], and layered splats for adaptive streaming judged on visual fidelity alone [47]. Only two of some thirty neural coding papers address progressive coding or bandwidth variation, and a doctoral plan still lists progressive transmission, quality assessment and error control as open [48].

Semantic codecs report the largest savings and the weakest comparisons. WiserVR sent 1.81 Mbit instead of 28.3 Mbit for 20 frames, in an AWGN simulation that excluded the delay of sharing the knowledge base [49]; learned point-cloud features gave more than 30$\times$ compression [50]; MagicStream reports up to 1195$\times$ less bandwidth [51]. The one generative prototype timed end to end cut bandwidth by 99.93% but raised latency from 980 ms to 13.66 s [52]. Fidelity is defined differently in each paper (PSNR on a noisy channel, joint-position error, text similarity), task metrics can disagree with perceived realism [53], and the baseline is usually BPG with LDPC rather than HEVC or VVC at matched quality. An order-of-magnitude saving is plausible; its price in inference delay and device energy is assumed rather than measured.

5.4 Holograms and touch as signals

Holographic and haptic content sit at opposite ends of the rate scale, and in both the requirement figures depend on an unstated representation. Akyildiz and Guo define a hologram as a recorded light field and then adopt point clouds as the working form [54]. Their worked example, a 1.5 m$\times$0.75 m light-field display with 0.5 mm hogels, needs 221 GB per frame, or 6.63 TBps at 30 fps, and they table holographic calls at 1–10 Tbps. Measured systems used 5–100 Mbps for compressed point clouds over live 5G [55], 173 Mbps for full-duplex light-field telepresence [56] and 60–200 Mbit/s for display-ready phase holograms [57]. The gap of four to five orders of magnitude separates uncompressed display arithmetic from compressed capture; the two are not estimates of one quantity. For live use, codec delay binds first: V-PCC needs 11–42 minutes to encode one second [58], and an uncompressed prototype ran at 8 fps with more than 300 ms of latency [59]. Optical holograms are only now becoming coded media, with a learnable JPEG-style codec that decodes in 4.2 ms [60] and

bit allocation between real and imaginary parts [61], both judged by PSNR or BD-rate alone.

Haptic signals need little rate and strict timing. Kinaesthetic samples are sent at 1 kHz or more to keep the control loop stable, and a 3-DoF packet carries 6 bytes of payload against 20–40 bytes of header [62]. Perceptual deadband coding sends a sample only when its change exceeds a Weber fraction (7–15% for force) and removes 80–90% of packets without perceptible loss; two independent implementations of the IEEE 1918.1.1 kinaesthetic codec reproduced the same rates and errors [21]. The standard separates delay-free kinaesthetic coding from vibrotactile coding, which may use block transforms with tens of milliseconds of algorithmic delay [63]; a speech-derived vibrotactile codec reached 2.3 kb/s without perceived distortion [6]. These results are established for a single contact point. Learned multichannel codecs report bitrate or traffic cuts of 80–92% [64, 65], mostly in simulation and against each other rather than a standard anchor, and the one standard-compliant encoder in the corpus still calls the standard future work [66]. Deadband coding also makes traffic bursty, so the periodic 1 ms flow that transport studies assume is not what a codec emits.

Open problems

P1. Semantic and generative codecs are not evaluated end to end. Of the semantic papers in the cell, one uses a real testbed and none reports a subjective test. Comparing them with HEVC, VVC and V-PCC at matched rated quality on an open 5G testbed, timing encoder and decoder inference, would show whether bandwidth savings survive latency.

P2. New representations have no perceptual ground truth. No public rating dataset exists for compressed Gaussian, neural or generative reconstructions viewed in a headset, so PSNR and LPIPS are unvalidated for their artefacts. A six-degrees-of-freedom rating study across codecs and rates would supply one.

P3. Geometry and views have never been compared, and MIV is almost untested on networks. One of about twenty MIV papers uses a real network. Encoding the same scenes as point clouds, meshes, MIV and Gaussian splats over replayed 5G traces, at matched rated quality, would show which representation minimises bitrate and latency.

P4. Codecs are tuned at fixed rates. Coding papers rarely use bandwidth traces, cellular links or 3GPP XR traffic models, and neural codecs lack bitrate ladders. Rate control driven by measured capacity, evaluated on frame-deadline misses, would test whether codec gains hold on mobile links.

P5. Hologram requirements describe no real hologram. Half the holographic papers name no representation, and no measured traffic trace exists for any of them. A benchmark that captures one scene, codes it as point clouds, splats, light fields and phase holograms, and records traces, would replace requirement tables with measured figures.

P6. Coded haptic streams lack shared content and loss tests with users. Haptic codecs are evaluated on one-off recordings, and none under realistic loss with participants. Released reference recordings and a within-subjects study of deadband and vibrotactile codecs over replayed 5G and Wi-Fi impairments would close both gaps.

6 Delivering Ahead of the User: Adaptation, Prediction and Caching

An immersive client displays only a fraction of what it could receive, and it must request that fraction before the user looks there. This section asks how a system decides what to fetch, at what quality and from where, ahead of the user's head and body (Figure 7). The function covers 1,079 papers (20.0% of the corpus), and its share fell from 27.5% in 2017–2019 to 17.4% in 2022–2024. Rate adaptation and adaptive streaming remains the largest problem cell in the atlas, with 788 papers, but it grew by 9% a year while the corpus grew by 21%. Viewport and motion prediction (130 papers) and caching and prefetching (161) are smaller and growing faster. The decline tracks that of 360-degree video, the content all three were built for, whose share fell from 36.3% to 18.8%. Evidence thins along the chain: 25% of rate-adaptation papers use a real testbed and 5% involve users; prediction is judged mainly on recorded traces (61%); caching is judged mainly in simulation (57%) and never with users. Codecs are treated in Section 5, remote rendering in Section 7.

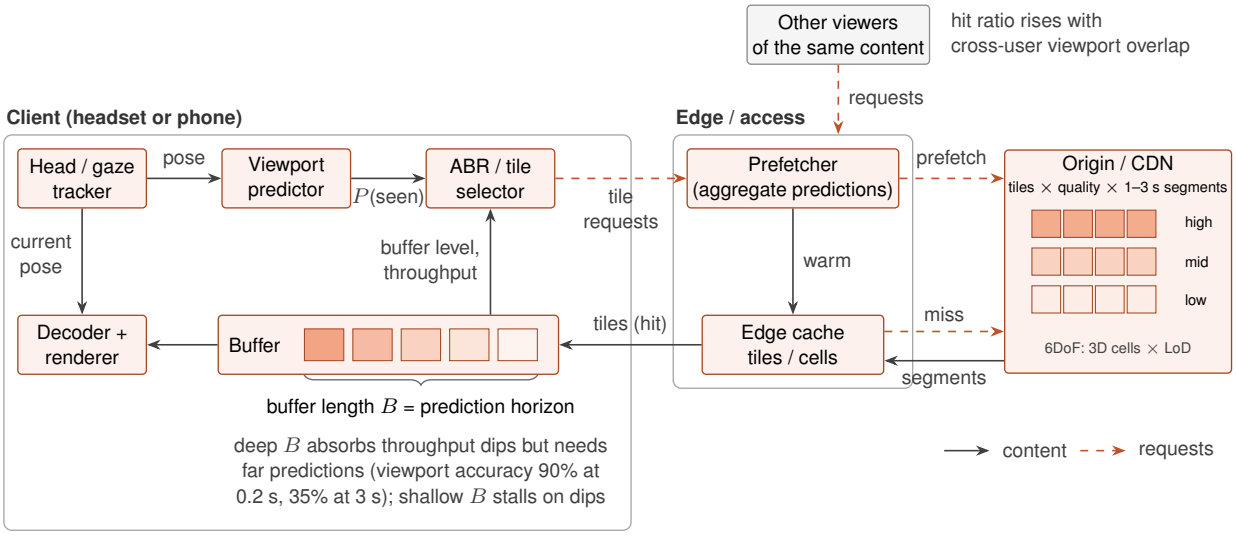


Figure 7: Viewport-adaptive delivery of tiled 360-degree or volumetric content across client, edge and origin. Each decision is a bet on where the user will look: the buffer length fixes how far ahead the predictor must see, and the edge cache pays off only when viewers' viewports overlap.

Table 2: Bandwidth saved by viewport- and visibility-adaptive delivery, ordered within each content type from idealised to realistic evaluation. The saving falls as the viewport and the network become real.

Content	Condition	Saving	Source
360-degree	Perfect knowledge of future motion (bound)	80%	[79]
360-degree	Scripted viewport, against all tiles at high quality	up to 72%	[80]
360-degree	HEVC tiles, static viewport / recorded head motion	65% / >40%	[25]
360-degree	Predicted motion with redundancy, <0.1% failure	45%	[79]
360-degree	Phones on commercial LTE, 4 users, 3 videos	26–35%	[16]
Volumetric	Gaze-driven delivery on HoloLens 2	up to 67%	[81]
Volumetric	Visibility-aware, commercial 5G, headset / phone users	40% / 10%	[18]

6.1 Viewport-adaptive streaming: spending bits where the user looks

Tiled delivery of 360-degree video saves bandwidth, and the saving shrinks as the test becomes realistic: from up to 72% with a scripted viewport to 26–35% for Flare on phones over commercial LTE (Table 2). The saving is paid for twice. Tiling costs coding efficiency, a median 7.3–10.4% for 1-s tiles [16] and up to 37.3% for an 8×5 grid [25]; decoding many tiles raised Flare's phone CPU load from 18% to 58% [16]. Prediction error is the main limiter, so controllers hedge: 360ProbDASH fetched tiles in proportion to their viewing probability and raised viewport PSNR by about 39% [67].

How the controller should be built is contested. Deep reinforcement learning, used by about a quarter of the cell, reports 20–30% gains over heuristics [68, 69]; BOLA360, a buffer-based controller with a near-optimality proof and no training, reports 6–110% [70]. The two families have never been compared on shared traces, and learned policies degrade on unseen viewing patterns and networks [71, 72]. Both optimise an assumed objective, a hand-weighted sum of viewport quality, rebuffering and switches. No abstract in the cell names an ITU-T QoE model, although tile extensions of P.1203 and P.1204.3 exist [73, 74], and a database of 378 viewport videos found that mainstream quality models fail for viewport-adaptive streams [75].

Most of this evidence comes from one on-demand client on an emulated link. Live streaming removes the pre-encoded ladder: YouTube Live delivered 360-degree streams with 4.2–44.2 s camera-to-display delay depending on mode [24]. Multicast exploits similar head motion across viewers and saved over 50% [76], but adaptation for several headsets behind one bottleneck has reached users in no study. Wireless access is the largest blind spot: only two abstracts in the cell use commercial cellular traces [18, 77], although mmWave variability already demands new mechanisms such as band switching [78].

6.2 Six degrees of freedom: adapting volumetric scenes

Volumetric streaming carries the idea into 6DoF, where the unit of adaptation becomes a 3D cell, an object or a level of detail, and visibility depends on position, distance and occlusion as well as direction. It is the fastest-growing part of rate adaptation. ViVo combined the three visibility tests and saved 40% of data on average, and up to 80%, over commercial 5G at SSIM above 0.99 [18]; the same content saved 40% for headset users but only 10% for phone users, so the gain depends on how people move. No general allocation policy has emerged. PCC-DASH found that the best multi-object strategy depends on the scene and the camera path [82]. The field assumes its navigation traces are representative, yet the newest public set has 15 participants [83], and about one volumetric adaptation paper in thirty evaluates with users.

Two-way volumetric communication breaks the buffer-based model altogether. Systems reuse low-latency DASH [84] or WebRTC [85]; a tiled user-adaptive point-cloud system needed 14 Mbps instead of about 300 Mbps in a 33-user study [86]. Adaptation in these systems is coupled loosely, if at all, to congestion control, and layered Gaussian-splatting representations are judged on visual fidelity rather than on behaviour under variable throughput [47].

6.3 Prediction: how far ahead, and against which baseline

The prediction horizon couples the two halves of the controller (Figure 7). A deep buffer absorbs throughput dips but forces the system to predict seconds ahead; a shallow buffer needs only short predictions but stalls when throughput drops. Accuracy falls steeply with horizon: Flare's best predictor reached median accuracies of 90.5%, 72.9%, 58.2% and 35.2% at 0.2, 0.5, 1 and 3 s [16], and a cross-dataset benchmark found no algorithm accurate much beyond 300 ms [87]. For on-demand content, earlier viewers are the strongest long-horizon signal: cross-user prediction raised accuracy by 20.2 percentage points [88]. Live content has no such history and relies on online learning or object semantics [89, 90].

Whether deep models beat simple predictors is the central controversy. On the Tsinghua dataset, persistence (keep the current viewport) reached a hit rate of 0.8355, against 0.8464 for an LSTM [91]; for 6DoF pose, simple statistical models performed comparably well [92]. Learned models report large gains mostly over other learned models. A fusion transformer reached an average precision of 0.954 at 1 s against 0.901 for the previous best, but on a random split that placed the same videos and users in training and test [93]. Fewer than one prediction paper in ten names a persistence, linear or Kalman baseline.

Two further assumptions go untested. Controllers need calibrated uncertainty to size tile margins, yet predictors output point estimates; predicting the predictor's own deviation saved 45% of bandwidth at under 0.1% failure [79], but no paper the atlas read reports calibration. Accuracy is also assumed to translate into experience, yet closed-loop tests with real systems fell from 9 of 74 prediction papers in 2016–2022 to 1 of 54 in 2023–2026. Pose prediction that hides motion-to-photon (MTP) latency is discussed in Section 7.

6.4 Caching and prefetching: placing content before it is requested

Viewport-aware caching beats popularity-only replacement, a result replicated by independent groups but always against weak baselines. Placing tiles by cross-user viewport similarity raised the edge hit ratio by at least 40% over LFU and 17% over LRU [94]. OpCASH covered more than 95% of the viewport from cache after 24 views and fetched 85% less data from the origin [95]. Prefetching extends to AR: world-scale AR retrieval missed 20% fewer deadlines with ten times fewer wasted bytes [96]. Joint designs that also transcode or render at the edge report lower latency in simulation [97, 98], a result that holds almost by construction, because the joint problem contains the decoupled one.

Three questions remain open. The first is granularity: fixed tiles, variable tiles that rarely recur

[95], or similarity caching over continuous orientations [99] have never been compared head to head. The second is the interaction with adaptation: cache hits distort the throughput estimates that DASH clients rely on [100], yet most caching papers assume an ideal client. The third is whether learned cache policies add anything beyond good viewport prediction, which cannot be settled while XRFlux remains the only shared benchmark [101]. Beneath all three lies the assumption that hit ratio stands for experience. No caching scheme in the atlas has been evaluated with users, and fewer than one caching paper in twenty addresses 6DoF content, which is reusable across viewpoints and users.

Open problems

P1. Prediction and adaptation are not validated in closed loop with users. Real-system tests of predictors fell to 1 of 54 papers after 2022, and ABR objectives are hand-weighted. A tiled player that runs persistence, Kalman and learned predictors under ABR policies over real bandwidth, with a within-subjects rating study, would show whether accuracy rankings survive.

P2. No dataset joins head motion, network conditions and ratings from the same session. Four dataset papers pair head data with network traces and four with ratings; replay removes the viewer's reaction to quality. Live sessions over Wi-Fi and commercial 5G, logging pose, packets and continuous ratings, would close this gap.

P3. Adaptation over real wireless access is under-evaluated. The cell has half the expected 5G papers and two abstracts with commercial cellular traces. Paired head-motion and radio traces from walking and seated users, replayed in emulation and run live, would test whether PHY and MAC hints reduce stalls.

P4. 6DoF adaptation and caching lack user-validated QoE and realistic navigation traces. Volumetric streaming grows fastest, yet trace sets have about 15 participants and volumetric caching is almost unstudied. Larger 6DoF traces, paired comparisons of rendered views and a level-of-detail edge cache shared by co-located users are needed.

P5. Caching gains have never been shown to users or outside simulation. No caching paper has users, and none of the coded, D2D or UAV caching papers uses traces or a testbed. An unmodified HTTP edge cache serving tiled DASH, comparing LRU, viewport-aware and Belady policies with viewers, would quantify perceptible benefit.

7 Where Computation Happens: Offloading, Remote Rendering and Latency

A headset has a few watts of power and a few milliseconds of slack. This section asks which stages of the sense–render–display pipeline should run on the device, at the edge or in the cloud, and how the delay that each split adds is budgeted, measured and hidden. The function holds 1,213 papers, 22.4% of the corpus, and its share has held steady (22.5% in 2017–2019, 22.1% in 2022–2024). It joins four cells: offloading and remote rendering (641 papers), latency (572, growing 25% a year against 21% for the corpus), rendered frames (484, whose share rose from 6.2% to 9.2%) and AR objects and overlays (592, whose share fell from 15.3% to 9.7%). The evidence is uneven. Offloading papers divide between real testbeds (37%) and simulation (34%), and only 4% involve users. Latency papers are more empirical (47% real testbed, 14% with users). Radio scheduling and transport are treated in Section 8, and QoE models in Section 10.

7.1 Choosing the split

Proximity is the one placement result that holds across methods. In a shipyard deployment, a fog gateway and a cloudlet both answered in about 1 ms, against 49.3 ms for a remote cloud; at 240 clients the fog gateway was four times slower than the cloudlet, and the Wi-Fi hop was excluded by design [104]. A simulation found that edge execution cut AR latency by 41.44% against local and 12.85% against cloud execution [105]. What leaves the device varies (Figure 8). VR systems

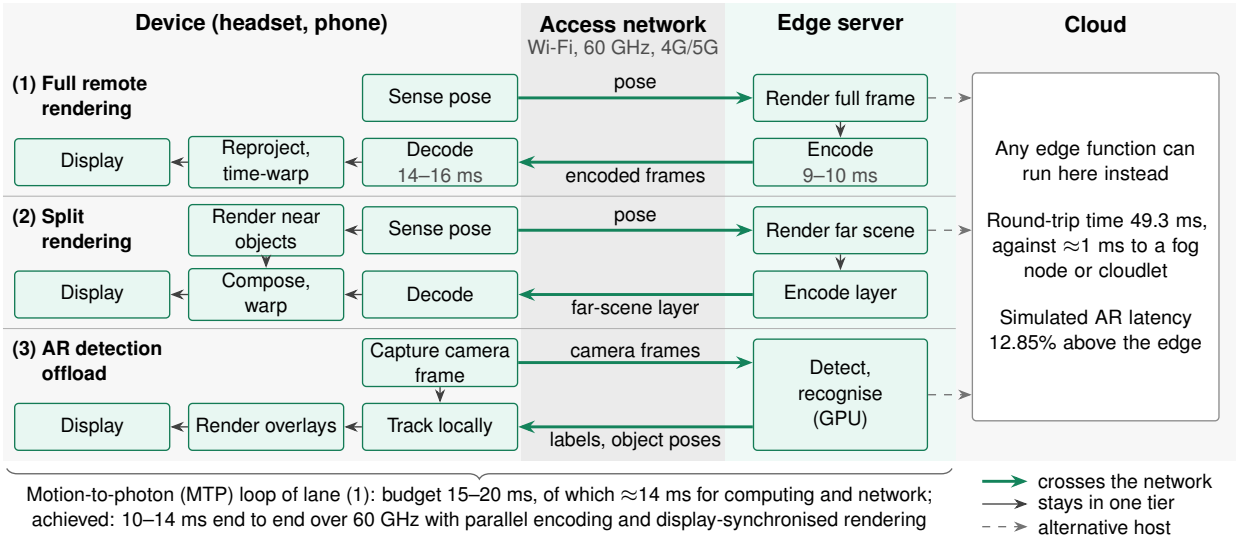


Figure 8: Offloading architecture: where each stage runs in full remote rendering, split rendering and AR detection offload. Each split is the point at which the pipeline crosses the access network; display, and in AR also tracking, stay on the device in every design. Values: MTP budget [1], 60 GHz end-to-end latency [102], encode and phone decode times [103], round-trip times [104], simulated edge advantage [105].

render remotely and stream frames. Split designs render near or interactive objects on the device and the rest remotely; one used 20–30% of the bandwidth of full remote rendering and tolerated delays of up to 200 ms [106–108]. AR systems offload perception and render locally. Neural scenes reopen the question: current systems stream pixels [109], dynamic Gaussians [110] or intermediate level-of-detail results [111], each against its own baseline.

Shared edges add contention, and here the evidence thins. Once many users offload, queueing at the edge server dominates [112, 113], yet schedulers for concurrent rendering sessions on one GPU exist only in simulation [114, 115]. The largest strand formulates offloading as joint computation–communication optimisation, solved by deep reinforcement learning [116] or Lyapunov control [117]. More than nine in ten learning-based controllers are evaluated only in simulation, and their inputs are assumed: a 2018 YOLO accuracy fit [118] reappears in later formulations [119, 120], and VR streams are modelled at a constant 25.32 Mbps [121].

Device energy, the main reason to offload, is contested. Models promise savings of 63% [122] and 85% [120]. Measurements disagree. Edge offloading halved smartphone energy for AR on average [123], and an edge VR testbed lowered client power [124]. A Quest Pro, however, drained its battery 15% faster with wireless offloading than with local rendering [125], and Air Link on a Quest 2 also cost more energy [126]. DeepDecision measured about 2900 mW for offloading regardless of bitrate and frame rate [127], against models that scale energy with transmitted data. Radio tail, decoder and display power, omitted from most models, are the likely cause.

7.2 Remote rendering: frames, codecs and pacing

Compression is mandatory, and its cost is set by pipelining and timing more than by link rate. Raw stereo frames at 2160×1200 and 90 Hz need 5.6 Gbps. Cutting the Cord split each eye's frame across four parallel H.264 encoders and aligned rendering with the client's display refresh; over 802.11ad it reached 10–14 ms end to end, against a sequential median above 26 ms, and cut missed refreshes from 5.3–14.3% to 0.1–0.2% [102]. Phones are tighter. Furion measured 9–10 ms to encode and 14–16 ms to decode H.264 frames, and 27% of the phone's CPU spent on packet processing at 400 Mbps [103]. Removing memory copies brought OpenUVR to 14.32 ms [128].

The renderer knows what the encoder must otherwise estimate, and using it pays. Object-aware quantisation with gaze saved 40–47% bandwidth in a subjective test [29]. Foveated shading alone did not reduce bitrate, while foveated post-processing or encoding did [129]. Other systems change

what is streamed: colour plus depth for client-side warping [130], object-space shading atlases [131], one eye rendered remotely and the other reprojected with 36% lower MTP [132], and pixel reuse across frames with 95% less frame data [133]. Each uses its own baseline, so they cannot be ranked (Section 5). Two assumptions remain untested. No corpus paper measures AV1 or VVC latency for rendered frames; the only mention is a position paper [134]. Frame sizes are bursty even under constant-bitrate coding [135], and on one headset the 3GPP traffic model underestimated effective bitrate by 25% [136].

7.3 AR offloading: stale results and the uplink

AR offloading succeeds only if the device hides the staleness of remote results. The replicated design runs a light tracker on the device between remote detections. In one system, without tracking, 16.7 ms of offload latency raised false detections from 0 to 31.56%; with tracking and region-of-interest encoding, offload latency fell from 34.56 to 17.23 ms on 2.4 GHz Wi-Fi [137]. CloudAR held 30 frames per second with at least 97% recognition accuracy [138], Jaguar reached about 33 ms end to end [139], and the pattern holds on HoloLens [140]. Once the server has a GPU, the network dominates: sending frames as H.264 video rather than images delivered about ten times more frames at 100 kbps [127].

The uplink is the bottleneck. In multi-user AR over LTE, median end-to-end latency was 12.5 s, the radio access network contributed 31.2%, and uplink bursts triggered TCP slow starts [141]. A 5G mmWave measurement found the uplink insufficient for baseline edge detection [142], while vision papers assume 5G solves the problem [5]. Result sharing among co-located users cut latency by up to 130× [112], but the overlap it relies on has never been measured. Targets are contested: below 20 ms [143], 50 ms [5] and 100 ms [140, 144] are all quoted. The right one depends on whether overlays are world-locked and tracked locally, and no study maps network delay to registration error.

7.4 Budgeting, measuring and hiding delay

Latency budgets are inherited rather than derived. The common figure is 15–20 ms MTP, leaving about 14 ms for computing and communication [1]; others assume 13 ms [4] or 5 ms [145]. Users are more tolerant once reprojection is active. CloudVR users did not notice 20 ms of added network latency [146], and ALVR users hardly noticed round-trip times up to 90 ms [147], whereas CloudXR quality fell from 75 ms [148]. In a crane simulator, display delays of 0–30 ms mattered while controller delay mattered only at 800 ms [149]. A budget therefore depends on which loop is meant and how delay is hidden, and offloading papers rarely state either. Measurement is cheap but rare: a photosensor rig [150] and a phone-camera method with under 10 ms uncertainty [151] measure MTP directly, yet fewer than one in twenty latency papers report jitter or tail latency.

Latency hiding is the established success. Post-render warping removed up to 80% of the aiming penalty of 80 ms added latency [152]. Rendering beyond the field of view, sized to the current latency, reduced delivery latency by about 22% in a simulation with real user data [153]. Motion predictors trained on more than 840,000 samples let the edge pre-render frames [154]. Reprojection has moved from rotation to translation [155], and selective time-warp leaves head-locked layers untouched [156]. Stalls of up to four frames at 90 Hz went unnoticed, and reprojection helped from eight frames upward [157]. Each method, however, is tested on its own motion data and with fixed delays, so the gains cannot be compared.

Open problems

P1. Learning-based offloading controllers are untested on real traces. More than nine in ten are evaluated in simulation with synthetic task sizes and channels. An open benchmark replaying measured render times, head motion and 5G/Wi-Fi traces under 3GPP XR traffic would show which gains survive.

P2. Device energy savings are modelled, not measured. One Quest Pro drained 15% faster with

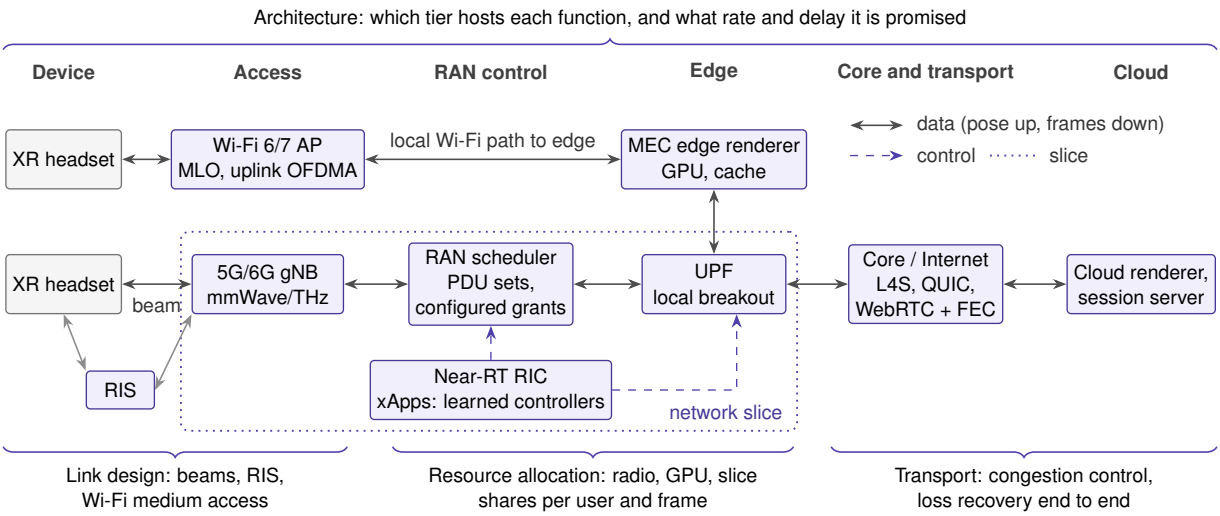


Figure 9: End-to-end 5G/6G and Wi-Fi delivery chain for XR, with the tier at which each strand of this section makes its decisions. Link design acts on the last hop, allocation inside the slice, and transport across the whole path; most work optimises one tier with the others held fixed.

offloading [125]; one edge testbed lowered power [124]. Component-level measurements of local and remote rendering across headsets, codecs and links would settle the direction.

P3. 3GPP XR traffic models go unused and unvalidated. No offloading paper evaluates against them, and the one empirical check found a 25% bitrate underestimate [136]. Commercial split-rendering traces fitted against the models are needed.

P4. Rendered-frame streams lack error resilience. Users rank loss above delay [147, 158], yet forward error correction, intra refresh and concealment for remote rendering barely appear in the corpus. A loss-trace study with user ratings would close this.

P5. No benchmark couples motion traces, network traces and MTP ground truth. Warping and prediction methods each use their own data and fixed delays. A shared dataset with photodiode MTP would let them be ranked.

P6. The AR camera uplink is unstudied at multi-user scale. Real-network evidence is single-user and pessimistic [141, 142]. A 5G testbed scheduling several AR uplinks in one cell would test whether edge AR scales.

8 Provisioning the Network: Radio, Resources, Transport and Architecture

This section asks how access networks, schedulers, transport protocols and system architectures supply the rate, delay and reliability that XR needs. Provisioning is the fastest-growing function in the corpus. It holds 1,187 papers (21.9%), and its share of output rose from 14.6% in 2017–19 to 27.2% in 2022–24, while the share of papers built on 6G rose from 0.6% to 9.4%. Three cells hold the work: resource allocation (645 papers), architecture and vision (380) and wireless link design (162). They differ sharply in evidence. 66% of allocation papers are simulations and 6% use a real testbed; 79% of architecture papers have no evaluation at all; link design is the most empirical (27% real testbeds). No paper in any of the three cells asked users what they perceived. The argument of this section is that provisioning research has solved well-posed optimisation problems whose inputs (QoE functions, rendering costs, traffic and requirement figures) were assumed rather than measured. Figure 9 places each strand's decisions in the delivery chain; offloading decisions are treated in Section 7.

8.1 Allocating radio, compute and money

Allocation divides spectrum, power, edge GPU time and caches among XR users so that each meets a delay or quality target. The established result is that these resources substitute for each other. Caching rendered fields of view and computing on the device can stand in for bandwidth [159, 160], and joint uplink, downlink and edge allocation saves mobile energy in collaborative AR [122]. Multicasting overlapping 360-degree viewports over mmWave raised HD delivery by 40–50% and cut 99th-percentile delay by 17–37% against unicast, in simulation driven by 50 real head traces [161]; the gain shrinks as viewports diverge, so it is unproven for interactive 6DoF content.

The objective is contested. The cell optimises at least four incompatible QoE constructs: a utility of tracking, processing and transmission delay [121], regressions of SSIM and VMAF [162], an attention-weighted "Meta-Immersion" score [163] and weighted sums of quality, delay and variation [164]. Gains of 20–37% are reported in each paper's own metric and cannot be compared, and none of the 137 allocation papers that optimise QoE was evaluated with users. Method is disputed too. Reinforcement-learning schedulers beat greedy, random and proportional-fair baselines [121, 165], while optimisation papers reach near-optimal decisions four orders of magnitude faster than a commercial solver [166]; the two camps rarely meet on the same traces.

Much of the rest is assumed. Rendering capacity is a free variable in joint radio–rendering allocation [163]; the only measured GPU figure is economic, a 4 GB vGPU slice that halved cloud-XR hourly cost while holding MOS at 4.38 or above [167]. Metaverse markets are the extreme case: a learned double Dutch auction reached social welfare within about 5% of the optimum [162], but every auction and incentive scheme in the cell uses synthetic valuations and arrivals. Standards-aligned work is more realistic about traffic and schedules PDU sets [168]. A delay-tracking PDU-set scheduler served 4–5 times more satisfied users than earlier PDU-set-aware schedulers under scarce resources [169], yet only 24 of the 282 reinforcement-learning allocators use 3GPP XR models. Real traffic undermines static provisioning: VR frame sizes fluctuate even under constant-bitrate encoding [135]. On O-RAN testbeds, latency-estimation control met 10–16.67 ms targets but over-provisioned [170], and a latency-steering xApp freed 15.8% more resources than an equivalent static slice [171].

8.2 The last hop: directional links, Wi-Fi and 5G

Link design must keep a multi-gigabit, low-delay channel to a moving head, and the tail, not the peak rate, is the limit. THz analysis gives 99.999% reliability at 20 ms with guaranteed line of sight, but under blockage reliability falls to 96% at 20 ms and tail delays reach 30–90 ms while the mean stays below 20 ms [172]. At 60 GHz, 3D head motion defeated beam search designed for 2D [173], and commercial hardware lost up to 4.72% of packets under 45°/s rotation [174]. Spatial diversity is the remedy that holds across studies. Dual-beam reception with coordinated access points raised reliability by up to 22.8% on real headset motion [175], distributed antennas kept 12 Gb/s under full rotation [176], and steering beams from head pose helps [177]: predictive beams from a neural radio field halved beam-search error on a 28 GHz array [178].

On Wi-Fi and 5G the bound is medium access, not physical rate. With frames aligned to the 802.11ad beacon interval, eight headsets at 400 Mbit/s met a 1 ms bound [179]. Wi-Fi 7 multi-link operation served 7 rather than 6 AR stations at 80 MHz [180]. Whether downlink or uplink binds depends on the assumed rates [180, 181]; on real hardware, uplink OFDMA with three headsets disrupted tracking [182]. 5G-Advanced link adaptation reports up to 67% more satisfied XR users [183], entirely in system-level simulation. Two technologies rest on assumption alone. RIS-assisted XR has never run on hardware, and RIS optimisers that take seconds per slot [184] must react within a head rotation [185]. No XR traffic has crossed a THz link; the analyses assume Poisson arrivals and fixed-size frames [172].

Table 3: Stated requirements against measured or derived figures. The gaps follow from unstated content representation, codec and control loop.

Quantity	Stated	Measured or derived
Per-user metaverse rate	100–250 Mbps [7, 192]; $\sim 10^3\times$ 5G throughput [191]	Horizon Workrooms 2–3 Mbps down, 0.6 Mbps up, +0.5 Mbps per extra user [193]
Holographic call	1–10 Tbps; 6.63 TBps for a light-field display [54]	5–100 Mbps for a compressed point cloud over live 5G [55]; <0.2 Mbps for a driven avatar [194]
Untethered VR link	15.75–26.25 Gbps uncompressed [174]	400 Mbit/s per 4K headset, eight per 802.11ad AP [179]
Interaction latency	1 ms in 58 abstracts, 2 with users	Haptic users tolerated 200 ms with adaptive control [195]; cloud VR RTT up to 90 ms hardly noticed [147]
Discomfort threshold	30 ms [192] vs <20 ms [14]	Not derived from a user study in either source

8.3 Transport over shared bottlenecks

Transport decides what reaches the headset when the bottleneck is shared and lossy. Its share of the corpus fell from 23.7% to 12.6% as delivery moved from buffered streaming to interactive rendering, although interactive traffic absorbs loss less well. Users confirm the risk. In cloud VR, RTT up to 90 ms was hardly noticed while loss of up to 8% mattered [147]; with CloudXR, QoE fell at RTT of 75 ms or loss above 6% (N=30) [148]; with Air Link, loss raised cybersickness more than delay did (N=16) [158]. Yet only 2 remote-rendering papers address forward error correction or error resilience [136, 186].

Congestion control for XR is nascent. L4S appears in 5 corpus papers. An L4S pipeline for volumetric conferencing converged 45% faster than Google congestion control and lost no frames where WebRTC lost 3% [187], and L4S has been carried over 5G for split rendering [188]; both used controlled bottlenecks with one competing scheme. Practice is ahead of research: Steam Link carries video and control in one QUIC tunnel [189], and WebRTC selective forwarding adapted 13–36% faster than TCP for volumetric streams in emulation [190].

8.4 Architectures and the figures they carry

Architecture papers decide where functions sit and state what the network must deliver. They agree on the qualitative list (high rate, low delay, reliability, interoperability) and on edge computing as a necessary tier [7, 14, 191]. The quantitative figures are where they fail. Table 3 sets stated requirements against measured ones; for rates the gap spans two to five orders of magnitude. The cause is unstated operating points: an uncompressed light field is not a compressed point cloud, and 360-degree video is not avatar state.

The figures propagate. One survey reproduces another's requirement table with a reliability target changed from $1 - 10^{-6}$ to $1 - 10^{-5}$ [7, 8], and its research agenda reproduces a third [196]; the apparent consensus is partly one source counted several times. Measured platforms tell a different story: throughput, latency and device load on five social VR platforms grew almost linearly with users, because servers forward avatar data directly [193, 197]. The scaling bottleneck is server logic, not access capacity.

Two architectural claims remain untested. Blockchain is placed in the interactive data path for ownership and trust [7, 196]; 44 architecture papers include a ledger, 41 of them without evaluation, and none has measured the delay it adds, although one concedes that replication raises delay with users [198]. Network slicing for XR, by contrast, has testbed evidence: AR, cloud gaming and video calls were measured under RAN-slice configurations on an O-RAN 5G network [199]. 6G visions rarely map their functions onto the 5G-Advanced XR mechanisms in Figure 9 [200, 201], so what

6G must add remains unstated.

Open problems

P1. Allocation objectives are not validated with users. None of the 137 QoE-driven allocators was evaluated with participants. A QoE-in-the-loop slicing study on an open 5G testbed, comparing static, latency-target and QoE-model xApps with ratings and sickness scores, would show which objective matters.

P2. Rendering costs in joint radio–GPU allocation are assumed. Profiling render and encode time and GPU power on edge hardware, then re-running published allocators under measured costs, would show whether their rankings survive.

P3. RIS and THz have never carried XR traffic. All RIS-for-XR work is simulated, and THz reliability rests on Poisson traffic. A pose-driven RIS prototype at 28 or 60 GHz, measured on frame deadline misses, would test the premise.

P4. Requirement figures are not derived from measured traffic. Stated per-user rates exceed measured ones by up to two orders of magnitude. A traffic atlas of shared-world platforms across user counts and avatar fidelities, compared with 3GPP models, would replace copied tables.

P5. Transport and loss resilience are untested on real bottlenecks. Users weigh loss above delay, yet L4S appears in 5 papers. Comparing L4S, delay-based and loss-based control with forward error correction over cellular and Wi-Fi links with cross traffic, reporting p99 frame latency, would fill this.

P6. Blockchain latency is uncosted. No study measures ledger delay against interaction deadlines. A benchmark of centralised, permissioned and rollup ledgers inside a live session would settle whether the ledger belongs in the data path.

9 Connecting People and Things: Shared Worlds, Touch and Telepresence

The previous sections treat one user and one stream. This section asks how networks keep several users, their bodies and remote machines consistent with each other. The function holds 814 papers (15.1% of the corpus), and its share of output fell from 19.1% in 2017–19 to 13.0% in 2022–24. Five cells make it up: multi-user synchronisation (349 papers, 6.5%), avatars and social state (347, 6.4%), tactile and haptic transport (344, 6.4%), holographic communication (121, 2.2%) and digital-twin data (319, 5.9%). Their paths diverge. Tactile transport fell from 9.8% to 4.4% of the corpus as the Tactile Internet agenda faded, while digital-twin content grew fastest, at 56% per year. Synchronisation and holographic papers are mostly built systems (58% and 56% on real testbeds), yet only 3% of synchronisation papers are controlled user studies. Haptic transport is the reverse: 25% simulation and 8% with users. Figure 10 places the five strands in one session and sorts their loops into three kinds: a control loop that can lose stability, interactive loops bounded by perception, and freshness loops bounded by the user's decision. Haptic and holographic coding belong to Section 5.

9.1 Shared state: consistency, scale and fairness

The design question is how much of the shared world each client must receive, and how fresh. Deployed platforms answer bluntly. Five commercial social VR platforms forward every avatar update unprocessed, so throughput, latency and client load rise roughly linearly with users; two-user end-to-end latency was 101.7–239.1 ms [197]. A controlled study located the limit in replication, not link capacity: update intervals grew from 18 ms with 2 clients to 158 ms with 125, and perceived fluidity collapsed beyond 25 avatars [207]. The largest shared world in the corpus had 225 clients [208], against metaverse definitions that demand thousands [7]. Interest management and dead reckoning, the classical remedies of networked virtual environments, appear in only 10 papers, none evaluated with users. This is the clearest case in the atlas of a problem solved in games that XR has

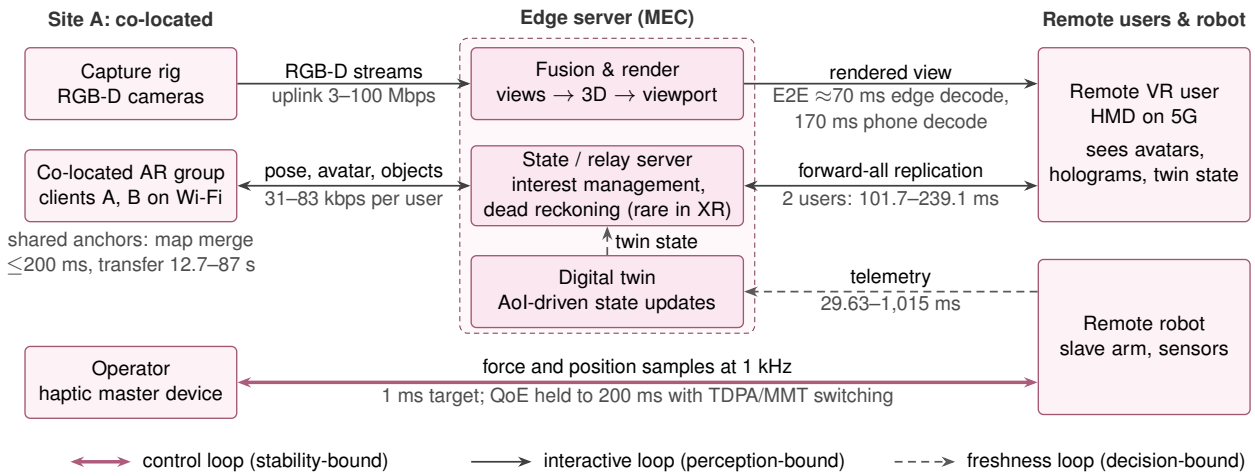


Figure 10: Architecture of a multi-party XR session: a co-located group and a remote user share state through an edge relay, a capture rig feeds edge fusion for holoportation, and an operator drives a remote robot whose digital twin updates every viewer. The loops differ in what bounds them (line style), so no single latency target serves the session. Numbers are measured values cited in the text: avatar rates and replication latency [197], holoportation rates and latency [55, 202], anchors [203, 204], twin updates [205, 206], haptic rate and tolerance [21, 62, 195].

not reused.

Consistency is traded against latency by assumption. Augmented Virtual Teleportation left its 1.2 s live video unsynchronised with avatar state to stay responsive [209], whereas two users actuating one digital twin without a policy conflicted in 33–56% of windows [210]. No study measures which inconsistencies users notice. In co-located AR the shared coordinate frame is the bottleneck: SPAR cut spatial inconsistency by up to 60% [211] and edge map merging finished within 200 ms [203], yet anchor transfers took 12.7–87.18 s [204]. Synchronisation faults are also the most common symptom in 2,649 bug reports from multi-user XR software [212].

Fairness is the most contested question. Of 26 delay-skew schedulers, 18 are simulations; they assume that equal delay means equal experience [213]. Predictive buffering cut inter-client skew by 76% for 100 clients, in a controlled testbed [214]. The user evidence points elsewhere: dyads perceived network impairment but completed tasks no slower [215]. The largest controlled study used 30 pairs [216], and a review of 56 studies found almost all optimise latency for an individual, not a group [217]. Group quality of experience (QoE) beyond two users is unmeasured.

9.2 Representing remote people: from pose streams to holoportation

Remote people travel as parameters, semantics or geometry, and the choice spans four orders of magnitude in rate. Parametric avatars are cheap: 24 B per update for a head pose, 216 B for a 17-joint skeleton [207], and 31–83 kbps for two users on VRChat, Rec Room and Hubs [197]. Expressiveness multiplies the rate: Horizon Worlds exceeded 4.5 Mbps downstream at 15 users [197]. Apple's spatial personas send semantic data at about 0.7 Mbps but lack rate adaptation and cap a call at five [218]. Geometry is costly. An operator trial streamed live point clouds at 5–100 Mbps, with end-to-end latency near 170 ms when phones decoded and 70 ms when the edge did [55]; depth packed into H.264 needed 3–30 Mbps [202]. Semantic drivers sit between: Mon3tr animated a Gaussian avatar with under 0.2 Mbps at about 80 ms [194].

Reusing the 2D real-time stack is established. A WebRTC selective forwarding unit for point clouds adapted faster than TCP [190], and L4S converged on bandwidth 45% faster than Google congestion control without frame loss [187]. Whether 3D beats 2D is contested. Project Starline [19] and stroke consultations with 30 outpatients [219] favour 3D; a floating-head prototype [220] and a VR2Gather–Zoom comparison [221] rated 3D below 2D video. Every one of these studies held the network fixed, although humanoid avatars lost more QoE under network distortion than

cartoon ones [222]. Two assumptions go unexamined. Semantic schemes report savings in PSNR, never against geometry at equal bitrate with users. Two-way calls are treated as downlink problems, yet only two papers measure the holographic uplink [223, 224], and the operator trial attached its camera by Ethernet [55].

9.3 Touch across the network: haptic loops and teleoperation

Haptic teleoperation is the only loop here that closes a control system, so delay threatens stability as well as experience. The field set its requirement early at 1 ms end-to-end latency with 99.999% reliability [62], and IEEE 1918.1 fixed a master–network–slave reference architecture [63]. The 1 ms target appears in 58 abstracts, only 2 of them evaluated with users. Measured tolerance is far looser and depends on the controller. Switching between time-domain passivity (TDPA) and model-mediated teleoperation (MMT) held QoE at delays up to 200 ms [195], and users noticed short haptic delays but compensated [225]. The requirement belongs to the controller, not to the network alone.

Compensation gives the clearest gains. Co-designing receiver-side prediction with radio resources saved up to 77.80% bandwidth [226], and octree-based MMT raised force accuracy by 44% with fewer than half the model updates [227]. Both rest on open-loop or synthetic trajectories; no predictor has been tested in closed loop with a live network, a stable controller and users. Transport results split by medium. A time-sensitive networking testbed kept high-priority flows at dedicated-wire quality under 200% over-saturation [228], whereas Wi-Fi needs MAC changes that reached a 3.72 ms worst case for Wi-Fi 7 only in simulation [229]. Haptic streams rarely travel with their video: no paper co-schedules them with remote-rendered frames. Cross-modal work even questions the usual priority of haptics, since deliberate haptic loss with visual restoration raised opinion scores by more than 80% at 30% loss [230]. The largest multi-user haptic study had six pairs [231].

9.4 Digital twins: keeping physical state fresh

A digital twin reverses telepresence: the network must keep a replica fresh enough for the user's decision. Two communities answer differently. Metaverse-framed work optimises age of information (AoI), latency and reliability for short packets [232]; optimal-transport association cut synchronisation time from 31.23–51.23 ms to 19.11–25.17 ms [233]. These studies assume their workloads, such as 10 Mb per twin update [233], and fewer than one in ten use a real testbed, against nearly two in three of other twin papers.

The prototype community reports what networks deliver. A robot vehicle mirrored on a headset updated in 29.63 ms over a local network [205], but middleware dominates elsewhere: 416 ms per transmission for a robot-arm VR twin [234] and 1,015 ms end to end for a cobot through a commercial IoT platform [206]. Every figure was called adequate. Compact state beats pixels: pose streaming kept remote work usable under added latency where video did not [235], and server push used three orders of magnitude less bandwidth than polling for a 5,005-building city twin [236]. Freshness metrics have multiplied, yet none has been tested against what a user perceives or decides.

Open problems

P1. Group QoE and fairness are unmeasured. 18 of 26 skew schedulers are simulated, and controlled studies stop at pairs [216]. Groups of two to four under per-user netem impairments, with a shared task score and perceived fairness, would give schedulers a validated target.

P2. No state-stream traces exist. Avatar, twin-update and haptic streams have no public traces or traffic models, so simulations assume sizes and arrival processes [226, 233]. Packet-level traces from instrumented open-source platforms, fitted to a 3GPP-style model, would make schemes comparable.

P3. Haptic tolerance is assumed, not measured. The 1 ms target persists while users tolerated 200 ms with adaptive control [195]. Factorial delay, loss and jitter studies across control schemes and XR tasks would yield tolerance curves for provisioning.

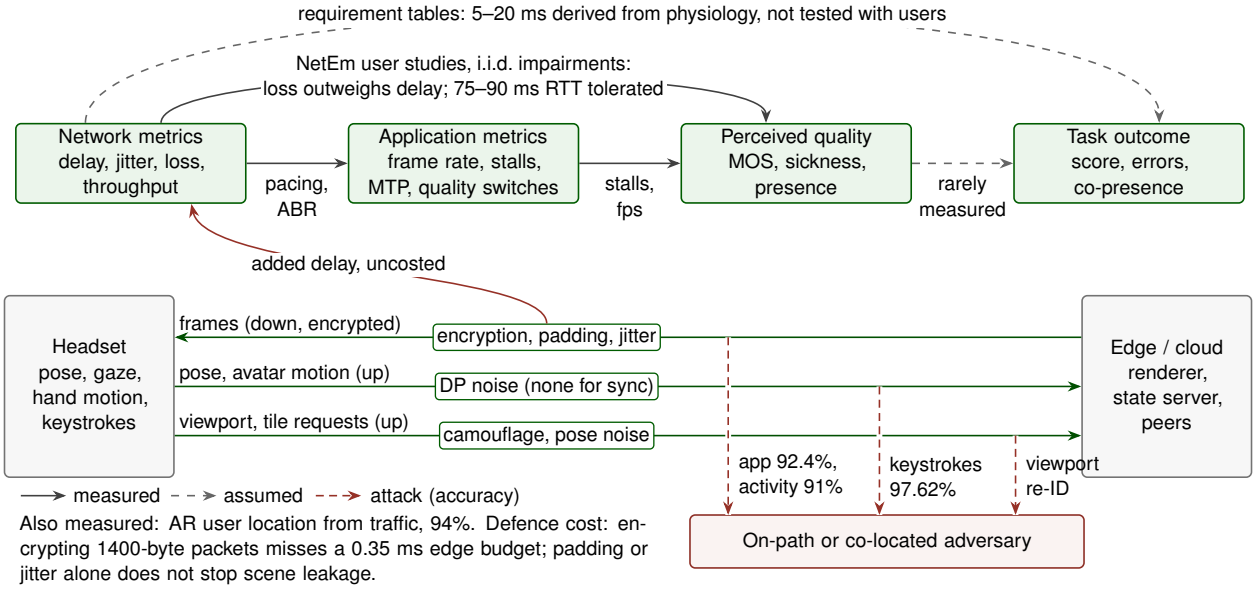


Figure 11: Measurement chain and attack surfaces of a networked XR session. Solid arrows are links that user or traffic studies have measured, dashed arrows links that are assumed; red taps show measured attack accuracies, and defences sit on the flows where they add delay that no testbed study has priced.

P4. Haptics travels without its video. No paper co-schedules haptic flows with remote-rendered frames. Joint scheduling on a shared Wi-Fi or 5G link, measuring haptic p99.9 delay, motion-to-photon latency and inter-stream skew, would close this gap.

P5. Holographic calls lack a conversational QoE model. All 3D-versus-2D studies hold the network fixed, and semantic avatars were never compared with geometry at equal bitrate. A two-way call study varying delay, loss and uplink bitrate against a matched 2D control is needed.

P6. Twin freshness is not tied to user decisions. AoI is optimised, yet no study links staleness to anomaly detection or control errors. Psychometric staleness curves for monitoring tasks would turn freshness budgets from guesses into requirements.

10 Assuring Experience and Trust: QoE, Security and Privacy

The previous sections optimise delivery against targets. This section asks how we know that a networked XR system is good enough for its users, and safe for them. The function covers 572 papers (10.6% of the corpus), and its share rose from 6.9% in 2017–2019 to 11.5% in 2022–2024. Two cells hold it, with opposite evidence profiles. QoE measurement and modelling (262 papers) is the most user-centred cell in the atlas: 39% of its papers test with users. Security, privacy and trust (310 papers) grew 48% a year, yet 39% of its papers have no evaluation and 2% test with users. The argument of this section is that both halves measure the ends of a chain but not its links. QoE studies relate network impairments to ratings under conditions that real links do not produce, and security studies measure attacks precisely but never price defences in delay or experience. Figure 11 draws the chain, the attack surfaces and the points where defences add delay.

10.1 What users weigh

Users of remote-rendered VR weigh loss above delay. On ALVR, round-trip times (RTT) up to 90 ms were hardly noticed, while 8% loss and a 35 Mbps cap degraded experience [147]. With CloudXR (N=30), QoE fell at RTT of 75 ms or more, at loss above 6%, or at 25 ms combined with 4% loss [148]. Loss also outweighed delay for presence and emotion on Quest 3 (N=16) [158]. In social XR, RTT up to 77 ms had no significant effect but jitter did [237]. The motion-to-photon (MTP) loop

explains part of this. Isolated stalls of up to four frames went unnoticed, and reprojection helped from eight frames (N=29) [157].

The latency budget is contested. Requirement papers set 5–10 ms RTT and 20 ms interaction latency from physiology [238, 239], a factor of four to ten below what users tolerated. Client-side reprojection could reconcile the two, but no study has varied RTT and latency compensation together. A second weakness concerns the impairments themselves. None of the 103 user-tested papers in the QoE cell replays measured network dynamics; all apply constant RTT, independent loss or throttling, and real 4G and 5G measurements enter only after the fact, as inputs to models fitted under constant impairments [240]. Wireless loss is bursty and correlated, so the thresholds may not transfer.

Other factors are content and group effects. For volumetric content, frame rate dominated in a 102-participant study [241], and two quality levels were indistinguishable at 5 m on a HoloLens 2 (N=32) [242]. Cybersickness is contested: fewer than 22% of viewers reported it for AR point clouds [242], and it was "not an issue" for 360-degree video [243]; content motion and display, not the network, seem to drive it. Group QoE is tested almost only in pairs [215, 222], although players blame the opponent for their own delay [244].

10.2 Models, standards and traffic

Objective models are established for coding impairments and unproven for network ones. PSNR and VMAF track MOS for 360-degree video [245, 246], colour PSNR correlated at 0.84 for point clouds [242], and the ITU-T P.1203 and P.1204.3 bitstream models have been extended to tiled streaming [73, 74]. Learned models fit well but locally. A cloud VR gaming model reached PLCC 0.93 and raised interaction quality by 1.20 MOS when used to adapt encoding [247, 248], and regressions on the CloudXR data reached a mean absolute error of 0.14 [240]; each was fitted and tested on a few contents. Temporal pooling is missing: P.919 uses 10 s stimuli while presence needs multi-minute tasks [249], and no model pools sub-second network events into session QoE. The gap reaches well beyond this cell. "QoE" appears in 811 abstracts, of which 117 were tested with users, and an atlas screen with two independent readers found 256 streamed-XR papers, mostly on resource allocation and latency, that lack an RTT, jitter and loss-to-MOS model yet could use one directly as objective, constraint or evaluator.

Network-side estimators are validated against formulas, not people. PERCEIVE predicted a parametric QoE index within 4%, but its authors state that the index was never checked against ratings [250]; data-plane estimators share this ground truth [251]. MetaVRadar classified metaverse activity from encrypted traffic with 92–100% true positives without measuring experience [252].

Traffic measurements show that the application stack, not the link, often sets what users see. WebRTC pacing in 5.56 ms batches set frame delay at 6–21 ms over Wi-Fi 6 with an RTT of 1.5 ms [253]. Under 27 Mbps throttling, adaptive bitrate kept cloud VR frame loss at or below 5.12% where a fixed 40 Mbps stream lost 32.51% [254]. Models fitted to one stack disagree: the 3GPP baseline underestimated effective VR bitrate by 25% [136], and the open RiftCat model matched mean but not 95th-percentile delay [255]. Shared worlds need far less than stated: Horizon Workrooms used 2–3 Mbps downlink [193], but loading avatar and world models congested shared networks as users joined [256].

10.3 Attacks on XR traffic

Encrypted XR traffic leaks what users do, and this result is established across platforms and groups. Traffic patterns of a mobile AR application located users with 94% accuracy [257], and ARSpy tracked users of multi-player AR in a field test with 12 volunteers [258]. On Meta Quest Pro, off-the-shelf classifiers identified applications with 92.4% accuracy and activities with 91%, from less than 10 minutes of traffic per application [259]; games were classified above 90% [260]. Offloading AR glasses reveal the type of physical scene through packet sizes and timing [261].

Motion data are the sharpest channel. Avatar motion received by an adversary's own client in

multi-user VR yielded 97.62% of typed keystrokes, and the attack generalised to four applications [262]; Wi-Fi channel state information yielded 69.75% [263]. The first result turns state synchronisation, which broadcasts head and hand poses to every peer, into a leakage channel. Platform audits add the provider as observer: about 70% of Oculus data flows were not properly disclosed [264], and child accounts reached advertising and tracking services that adult accounts did not [265].

10.4 Defences and their uncosted latency

Defences are proposed far more often than they are measured. Of 286 corpus papers that mention both a security term and latency or QoE, 40% have no evaluation and 6 were tested with users. Authentication schemes for metaverse avatars and tele-surgery are proven in BAN logic, AVISPA or Real-or-Random models and costed in operation counts [266, 267]; cross-metaverse schemes [268] are not measured during the hand-off where re-authentication falls. "Metaverse" intrusion detectors report 96–99% accuracy on IoT and enterprise datasets that contain no XR flows [269, 270]. The exceptions are line-rate detection in Tofino switches at up to 99% accuracy with under 5% of switch resources [271] and the first labelled metaverse attack dataset [272]. Blockchain is contested: a 10-user pilot kept 88 FPS at 142 transactions per second [273], while zero-knowledge proof generation dominated overhead in a semantic-communication defence [274]. Federated learning failed for VR biometric authentication, below 10% accuracy [275].

The privacy–QoE trade-off is modelled, not perceived. Viewpoint leakage persists under federated prediction once prediction errors or QoE are uploaded [276]. Noise remapping reduced the QoE loss of privacy noise by 55–100%, to 0–3% at maximum privacy [277], and client-side pose noise lowered re-identification on real 6DoF traces [278]; none of the 15 papers in this line asked users. Selective attribute-based encryption cut encryption and decryption time by up to 50% and 80% and kept volumetric content cacheable [279]. The transport cost is known only at its extremes. Encrypting 1400-byte packets did not fit a 0.35 ms edge budget [280]; secured VR streaming under a flood held about 30–32 ms in simulation [281]; and padding or jitter alone did not stop scene leakage, while combined defences conflicted with QoS [261]. No real-testbed paper reports a defence's latency together with QoE or frame rate. A defence should be reported as a point on a privacy–QoE plane: attacker accuracy against added p99 frame latency and bandwidth, on a real split-rendering pipeline.

Open problems

P1. Network-side QoE estimators are not validated against users. Estimators predict parametric indices, and none targets interactive remote rendering. A public benchmark pairing encrypted packet traces, client frame logs and per-minute ratings for cloud VR ($N \geq 30$) would let estimators be trained and compared.

P2. XR traffic models are open loop. Each is fitted to one stack, one user and short captures, without head motion or rate adaptation, and errs in the tail. A closed-loop generator from several stacks and users, validated against p95 frame delay with 1–12 real headsets, would replace them.

P3. Defences are not costed against the frame deadline. No testbed study reports a defence's latency with QoE or frame rate. Measuring TLS, DTLS and QUIC with padding, bucketing and jitter on split rendering, as attacker accuracy against p99 frame latency, would give providers a price.

P4. Authentication is validated symbolically, never during hand-off. Cross-platform schemes have no measurements. Implementing published schemes and 5G re-authentication on a headset–edge testbed, and counting frames delayed per hand-off, would show which fit the interaction path.

P5. Intrusion detectors are trained on traffic without XR flows. One labelled XR attack dataset exists. Capturing benign and attack XR traffic on an open 5G testbed and testing published detectors across datasets would show whether 96–99% accuracies transfer.

P6. Viewport and pose privacy has no user validation. QoE losses come from prediction accuracy. A within-subjects study comparing noise, camouflage and client-side prediction at equal re-identification risk, with MOS and sickness scores, would show what users notice.

Table 4: Nine cross-cutting themes. Gaps: verified cell gaps that instance the theme (high-confidence in brackets). Cells: atlas cells in which the theme recurs. A gap can belong to more than one theme.

	Theme	Gaps	Cells	Indicative evidence
Evidence foundations				
T1	No open traces, datasets or benchmarks	26 (17)	19	37 dataset papers; 3.9% of abstracts mention released code or data
T2	Evaluation stops at the simulator	32 (15)	19	7% of 812 learning papers tested on a real network
T3	Standards are not the evaluation substrate	17 (9)	15	63 XR abstracts name 3GPP; none evaluated with users
Objectives				
T4	Network metrics not linked to experience	34 (20)	20	811 abstracts invoke QoE; 14% test it with users
T5	Requirement figures inherited, not derived	11 (5)	9	1 ms target in 58 abstracts, 2 tested with users
Blind spots				
T6	Uplink and sending side neglected	7 (1)	6	"uplink" in 2.1% of abstracts, 71% of them simulated
T7	Multi-user scale untested	19 (4)	14	largest shared-world test: 225 clients [208]
T8	Security and privacy not costed	15 (5)	8	6 of 286 security–latency papers tested with users
T9	Transport under real impairments	11 (2)	11	5 corpus papers apply L4S

11 Cross-Cutting Gaps

Read side by side, the open problems of the six functions repeat themselves in different vocabularies. Take a viewport predictor never tested in a closed loop, an offloading controller that never left the simulator, and an allocation scheme whose QoE objective was never shown to a user. They are three instances of one shortfall. The atlas verified 159 gaps across its 22 cells, 57 of them with high confidence. Nine themes account for 132 of them (83%), including 53 of the high-confidence ones (Table 4). A theme was kept only if it held gaps from at least six cells. We group the nine themes into three families.

11.1 Evidence foundations

The first family concerns what claims are tested against. *Data are missing* (T1). There are no public traffic traces for AR offloading, avatar streams, twin-state updates, holographic calls or haptic flows. No 360-degree dataset records head motion, network conditions and ratings in the same session, and no latency benchmark couples motion, network delay and photodiode-measured MTP. Even simple baselines are often missing: a persistence predictor reached a hit rate of 0.8355 against 0.8402 for a sequence-to-sequence model on the same data [91]. *Evaluation stops at the simulator* (T2). Whole technology lines, including reconfigurable surfaces carrying XR, terahertz XR links and metaverse resource markets, have never met hardware. Where learned controllers were tested on unseen users or content, they degraded [71, 72]. *Standards are not used as the substrate* (T3). 3GPP XR traffic models appear almost only in radio simulations. When one was checked against measurement, it underestimated cloud-VR bitrate by 25% [136]. OMAF, MPEG Immersive Video, IEEE 1918.1.1 and ITU-T P.1203/P.1204 are rarely the baseline configuration. The three gaps share a cause. Collecting traces and building testbeds is costly, and venues that publish algorithms rarely credit either.

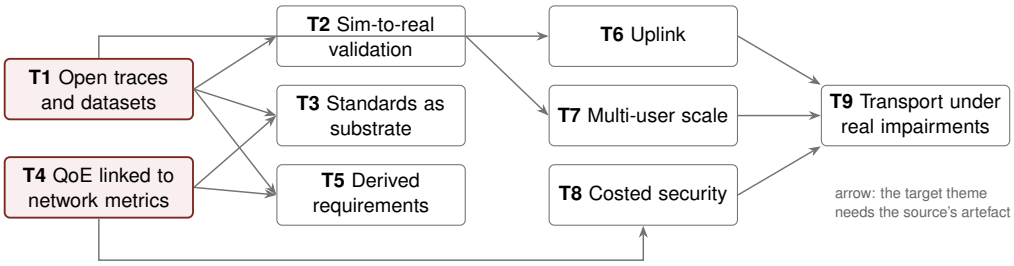


Figure 12: Dependencies among the nine themes. Open traces (T1) and a validated mapping from network metrics to experience (T4) are the roots. Without them, simulations cannot be validated, requirements cannot be derived, and allocation, transport and security results cannot be compared in units that matter to users.

11.2 Objectives

The second family concerns what is optimised. *Network metrics are not linked to experience* (T4). This is the largest theme. Most papers that optimise "QoE" optimise a formula whose weights were chosen rather than fitted. Where users were asked, the answers diverged from the proxies. Packet loss outweighed delay in VR cloud gaming [158]. Users of collaborative VR could not tell forms of latency apart [215]. CloudVR users did not notice 20 ms of added latency [146], while offloading papers optimise to budgets of 5–13 ms. *Requirements are inherited* (T5). Vision papers state 100–250 Mbps per metaverse user [7, 192], while a measured commercial social-VR platform used 2–3 Mbps downstream [193]. Holographic requirements range from 5–100 Mbps for compressed calls [55] to 6.63 TBps [54]. Haptic users tolerated up to 200 ms with adaptive control [195], against the 1 ms figure that motivates much tactile-internet work. A requirement that names neither the representation, the codec, the loop nor the percentile cannot be checked.

11.3 Blind spots

The third family concerns what is left out. XR sends a continuous upstream, yet the *uplink* (T6) is studied far less than the downlink, and the measured cases are pessimistic. On public LTE, multi-user AR had a median end-to-end latency of 12.5 s, with the radio access network contributing 31.2% [141]. With three headsets on Wi-Fi 6, uplink OFDMA delayed tracking data [182]. *Scale* (T7) is untested. Measured social platforms scale linearly because they forward all avatar data [197], and controlled QoE studies use pairs of users. *Security is not costed* (T8). Attacks are measured and effective: keystrokes were inferred from avatar motion with 97.62% accuracy [262]. Defences, however, are seldom priced in frame latency. *Transport* (T9) has not followed the move from buffered streaming to interactive delivery. Users rank loss above delay [147, 158], yet forward error correction and L4S are rare. An L4S pipeline for volumetric conferencing converged 45% faster than Google Congestion Control [187], but in a controlled bottleneck.

11.4 How the themes depend on each other

The themes are not independent (Figure 12). Two are roots. Without open traces (T1), simulations cannot be validated (T2), requirements cannot be derived (T5), standard traffic models cannot be checked (T3), and uplink and multi-user workloads cannot even be described (T6, T7). Without a validated mapping from network metrics to experience (T4), results on allocation, transport and security (T8, T9) cannot be compared in units that matter to users. The two smaller clusters outside the themes point the same way. Device-energy savings from offloading rest on models, and the few measurements disagree: one standalone headset drained its battery 15% faster when offloading wirelessly [125], while an edge VR testbed found lower client power [124]. Neural scene representations have no tested rate adaptation, cache or split point. Both clusters need the same two roots. This ordering shapes the agenda that follows.

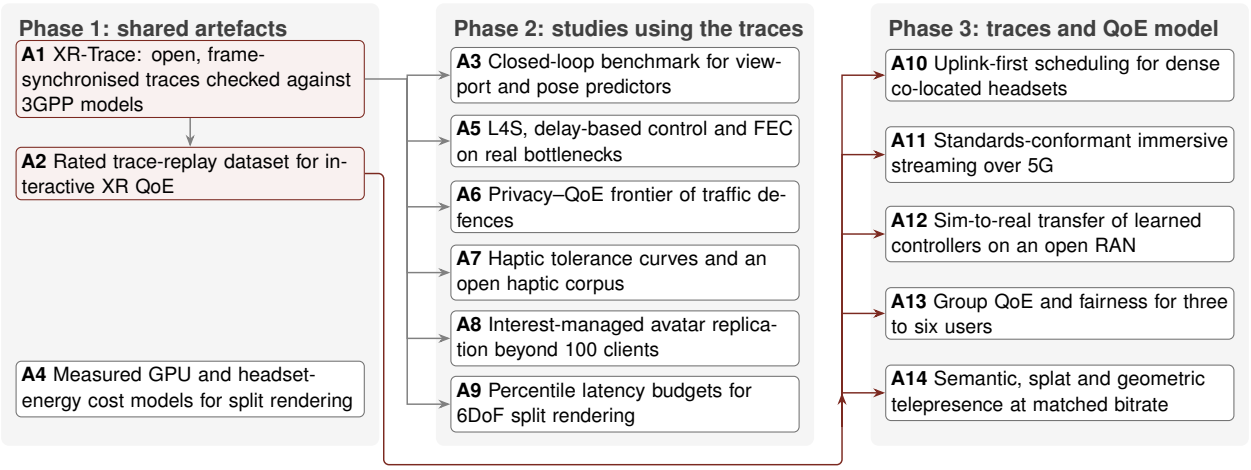


Figure 13: Research agenda as a dependency roadmap. Phase-1 projects (red) build the artefacts that the others consume. A1 supplies traces to every later project (grey arrows); A2, built on A1, adds the validated QoE model that the Phase-3 projects optimise against (red arrows). A4 stands alone.

12 Research Agenda

The open problems of Sections 5–10 merge into 14 projects that a university networking or multimedia laboratory could start now. Together they would close 87 of the 159 verified gaps, 38 of them with high confidence. We order them by dependency, following Figure 12: projects that produce shared artefacts come first, because the others consume them (Figure 13).

Phase 1: build what others need. *XR-Trace* (A1) would instrument the main workloads on 5G standalone and Wi-Fi 6E/7, with one, four and eight users. These are split-rendered VR, AR camera offloading, avatar and shared-state traffic, tiled 360-degree video and twin-state updates. Each trace would record pose, encoder output, packets at both ends, application events and photodiode MTP on a common clock. The project would then fit traffic models, compare them with the 3GPP XR models, and release the traces with a replay harness for ns-3 and NetEm. It closes 11 gaps and ranks first because 11 projects consume it. The *rated replay dataset* (A2) would have participants perform interactive tasks while the network replays A1's traces. It would collect momentary and session ratings, sickness scores, task performance and packet logs together, so that a QoE model can be fitted and then used as the objective by other work. It would also measure the difference between replayed dynamics and independent impairments of the same mean. *Energy cost models* (A4) would replace the assumed GPU and battery costs inside offloading optimisers with measured ones.

Phase 2 and 3: answer the open questions on shared ground. With traces in hand, five studies become cheap and comparable. They are a closed-loop predictor benchmark with persistence and Kalman baselines (A3), transport on real bottlenecks (A5), a privacy–QoE frontier for padding and jitter defences (A6), haptic tolerance curves (A7) and interest-managed replication beyond one hundred clients (A8). A percentile latency budget for 6DoF split rendering (A9) completes the phase. With a validated QoE model as well, the remaining projects can optimise against experience rather than a proxy. These are uplink-first scheduling (A10), a standards-conformant 5G pipeline (A11), sim-to-real transfer of learned controllers as xApps on an open RAN (A12), group QoE and fairness (A13), and a matched-bitrate comparison of semantic, Gaussian-splat and geometric telepresence (A14).

The evaluation bar. Each project should meet the standard that Transactions-level reviewers apply and that most of the corpus does not meet. First, a *baseline ladder*: a trivial baseline, a classical method, the strongest recent method re-implemented under the same conditions, an ablation and,

where computable, an oracle. Second, *realism with control*: real hardware for the claim and trace replay for repeatability, with the gap between simulation and measurement reported. Third, *out-of-distribution tests* for anything learned, on held-out users, content and networks. Fourth, *sound statistics*: repeated runs with confidence intervals and, for user studies, a power analysis, within-subjects designs and mixed-effects models. Fifth, *reported cost and released artefacts*: run time, energy and signalling overhead on the target device, with code, traces and analysis scripts published.

13 Conclusion

Networked XR has grown almost fourfold in eight years and has changed character on the way. It began as a multimedia-systems field centred on 360-degree streaming. Its claims were tested with real players, real traces and, often, real viewers. Today it is dominated by resource allocation, edge computing and 6G architecture. It models XR traffic generically and tests most new claims in simulation. The empirical core has not disappeared, but it has not grown with the rest. This core covers latency measurement, multi-user and holographic systems, and QoE studies.

Organising the field by function makes the consequences visible. The functions that build systems, compute and connect, rest on real testbeds in 42% and 45% of their papers. Provisioning, now the largest and fastest-growing function, does so in 11%. Across all six functions the same pattern recurs. Optimisers are well posed, but their inputs are assumed rather than measured. The QoE they maximise was never shown to users. The requirements they meet were copied from earlier surveys. The traffic they carry was synthesised.

The remedy is less a new algorithm than shared ground: open, frame-synchronised traces of real XR workloads, and a QoE model fitted to people experiencing replayed network dynamics. With these two artefacts, the questions that recur in every section become answerable on common terms: whether a learned scheduler beats a heuristic on a real cell, how much latency a defence costs, and what a hologram really needs. Without them, the field will keep growing faster than its evidence.

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